Framework for Simulation-Based Assessment of First- and Last-Mile Fleet Sizing Strategies for a Multimodal Transit Interchange

Case study : Yeshwanthpur Railway Station, Bengaluru

M.Tech thesis report

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Abstract

First- and last-mile (FLM) connectivity remains one of the most persistent barriers to public transit adoption in Indian cities. At major railway interchange hubs, the absence of a structured framework for sizing and managing FLM vehicle fleets leads to excessive waiting times and over-reliance on informal para-transit. This thesis develops an agent-based simulation framework for assessing FLM fleet sizing strategies at a multimodal transit interchange, demonstrated through a case study at Yeshwantpur Railway Station, Bengaluru.

The framework is implemented in AnyLogic and models first-mile commuters and last-mile passengers as heterogeneous agents whose mode choice across five modes — Walking, Auto-Rickshaw, BMTC bus, Namma Metro, and Private vehicle — is governed by a Multinomial Logit (MNL) model. Socioeconomic attributes including income group, gender, luggage weight, vehicle ownership, and group size are assigned stochastically, and utilities are computed from travel time, travel cost, and waiting time. Demand is calibrated from one and a half months of IRCTC reservation chart data covering approximately 25,000–33,000 daily passenger movements at Yeshwantpur, and network inputs include real BMTC GTFS schedules and Namma Metro Green Line station data. The simulation is evaluated across four fleet sizes ($F = 300, 400, 500, 600$) under a base case, a policy case of doubled bus frequency, and a sensitivity analysis on key MNL coefficients.

Auto-rickshaw dominates last-mile trips at 42–43% mode share under the base case, with bus fallback rates persistently at 38–42% across all fleet sizes, driven by temporal service gaps rather than traveller preference. Doubling bus frequency reduces average bus waiting time by 2–3 minutes but produces negligible mode shift, as last-mile passengers face a structural constraint at the four key interchange bus stops that frequency alone cannot resolve. The sensitivity analysis identifies β_{OVTT} as the dominant parameter: heightened waiting time sensitivity drops last-mile bus share from 38% to 27%, confirming that reducing actual waiting time is a more effective policy lever than adjusting vehicle or group preferences. The framework shows that fleet size directly affects how long passengers wait for an auto-rickshaw, with larger fleets cutting peak delays significantly. Pinning down a single best fleet size, however, requires first agreeing on an acceptable waiting time threshold — and this framework is built to support exactly that kind of analysis. The key contribution of this study is a replicable simulation framework that brings out the link between auto fleet sizing and bus scheduling — two decisions that are usually made independently, but together shape how well a multimodal interchange actually serves its users.

Keywords: first-mile last-mile connectivity, agent-based simulation, fleet sizing, mode choice, multinomial logit, multimodal interchange, GTFS, Yeshwantpur, Bengaluru, auto-rickshaw, public transit

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List of Abbreviations

ABM Agent-Based Modeling. 13

AVs Autonomous Vehicles. 17

BMRCL Bangalore Metro Rail Corporation Limited. 39

BRTS Bus Rapid Transit System. 8

DRT Demand Responsive Transit. 13

DSRT Demand Responsive Shared Transport. 13

FLM The First and Last Mile. 4, 12, 18

FM First Mile. 7

INR Indian Rupees. 50

IPT Intermediate Para-Transit. 8

MNL Multinomial Logit Model. 19

NL Nested Logit. 16, 19, 22

NMT Non-Motorized Transport. 7

OVT Out-of-vehicle time. 17

PCFI Per-Capita Family Income. 19

RMP-2031 Revised Master Plan for Bengaluru – 2031. 9

YPR Yeshwantpur Railway Station. 9, 34, 39

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Chapter 1

Introduction

1.1 Background

How often we talk about traffic in Bengaluru and in that context we always bring the concept of moving the people not the vehicle. Karnataka Government most often came up with a solution to built more and more roads for example recent addition of double decker and tunnel road but which will create induced demand and influence people to use more private vehicles also to own one if they have not. Apart from that these flyovers and tunnel roads will also act as a shifting of bottlenecks(P. A. Verma *et al.* 2024).
On the other hand, Researchers have always came up with the concept of prioritizing the public transit over the private vehicles as these laid the foundation to sustainable urban mobility by reducing vehicle ownership and alleviating congestion (Vuchic 2017; D. He *et al.* 2013). But, it has always been observed that the major problem with transit is its first and last mile connectivity.
Let's discuss a little bit about that concept. Whenever we are going to use transit for our trip, our entire journey decomposed into three parts and these are primary leg, main mode and the secondary leg (refer figure 1.1). Let's talk about them one by one-

The primary leg (FM): It is also known as the First Mile (FM) leg of the journey. It is the beginning of the trip and in this leg commuters have to reach to its nearest transit stop from their origin to start its main leg which is usually performed by NMT, IPT or private vehicles (either two wheelers or four wheelers). Now, let's discuss little bit about NMT and IPT-

- **NMT :** It stands for Non-Motorized Transport (NMT) and which incorporates walking and cycling and also sometimes referred to as active mode of

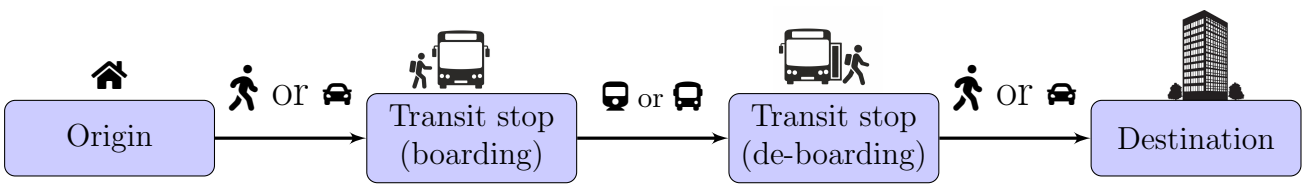


Figure 1.1: Typical line diagram of different legs of a journey covered through public transit

transportation.

- **IPT :** It stands for Intermediate Para-Transit (IPT) and which incorporates autos, taxis and ride hailing services like OLA, UBER, Rapido etc..

The main or line-haul leg : It is the longest part of the journey both in terms of distance and as well as time and mostly it is covered through Bus, Metro rail, Sub-urban rail , Bus Rapid Transit System (BRTS) etc..

The secondary leg (LM) : It is also known as the last mile leg (LM) of the journey and which takes the commuters to its final destination after deboarding from its main mode. Similar, to primary leg here also commuter can accomplish their journey through NMT, IPT or private vehicles.

1.1.1 Existing transport scenario in Bengaluru

Urban centres in India currently house about 31% of the population, projected to rise to 40% by 2031 (Sankhe *et al.* 2010).
The traffic forecast indicates 3 time increase in vehicular traffic than the present status, resulting in high traffic congestion, economic and environment losses for example the total emissions in 2015 due to vehicular traffic is estimated at 196 tonnes of CO per day, 50 tonnes of HC per day, 9 tonnes of SPM per day and 321 tonnes of NO_x per day (Central Pollution Control Board 2021). The Public Transport share is expected to decrease to 36% from the present 47% even if the bus fleet is doubled. The current bus fleet size across the city is around 7038 including EV (Bangalore Metropolitan Transport Corporation (BMTC) 2024) and is expected to rise to 15000 (Bangalore Development Authority 2017) and that is a huge number along with the private vehicles and auto and taxi services which comes from autonomous body as well as from the ride hailing services. Add to that a significant nos of two wheelers which puts a huge nos of vehicles into the road network on regular basis. The road network has its limited capacity and because of this huge nos of traffic most of the major corridors exceed its capacity (Bangalore Development Authority 2020). Since, Bengaluru has grown as a mutation corridors i.e. its residential areas also has some commercial activities (Rites 2019) and the uncontrolled growth of the city it is impossible to expand the road network capacity of the city.

On other other hand, because of huge nos of vehicles on the network, average vehicle speed on along the network reduces drastically and along the most of the major corridors its less than 8 km/hr and during peak period which becomes less than 5 km/hr just worsening the condition more (Bangalore Development Authority 2017). Because of that people of Bengaluru got stuck on the traffic more often and studies claim that its average value reads around 1hr per day (Bangalore Development Authority 2020) leads to reduction of productivity of people and the city .

Because all of these situations mention above the transport system of Bengaluru is characterised by a high dependence on private motorised modes, particularly two-wheelers and cars, accompanied by declining average travel speeds and increasing congestion levels (Bangalore Development Authority 2017).

1.2 Motivation

For the revival of traffic network one of the major approach cited in Revised Master Plan for Bengaluru – 2031 (RMP-2031) is to integrate intermodal hubs where different mass transit corridors meet (two or more public transit corridors). RMP 2031 has proposed 36 intermodal interchange stations (Bangalore Development Authority 2017) and Yeshwantpur being one of the major intermodal hubs which currently has

- Indian railway station
- BMTC bus stops all around the hub
- BMRCL metro station

and also upcoming Corridor 1 and Corridior 2 of Bengaluru Sub-urban platform.

So it will become a proper hub having all available modes of transportation in Bengaluru. But for the time being if we only consider the Yeshwantpur Railway Station (YPR) station in which many long distances trains terminates or starts or passes through along with that same goes for intercity trains. Whenever these long distance trains starts or terminates at those stations every time nearly around 1000 people (sometimes more than that) came into the system who needs FMLM services. And if we consider in a span of 2 to 3 hours during the peak period there might be 8 to 10 trains which are either starting or terminating at those stations making a demand of around 10 to 15 thousands people to access that station area. Along with that intercity trains keep on coming in between and they also generate some demands and to worsen the condition, Indian Railways normal day to day operation of getting late and what if there is 2 to 3 off peak period trains also getting into the mix and all of a sudden demand goes to near about 25 thousands people within a span of three hours. Even if we consider 10% of these commuters are needed FMLM services and by considering average occupancy level of the IPT's in Bengaluru i.e. around 1.7 (Bangalore Development Authority 2020) then also there is a requirement of around 1300 such vehicles accessing a small part of the network and make it heavily congested.

Thus, it necessitates to consider such a case and came up with a framework how we can reduce that number and re-route those vehicles effectively to keep the commuters delay within their tolerance limit and maximize the systems efficiency.

But one of the major challenge in integrating the different modes and promote public transit as the dominant mode is their first and last mile connectivity (Araghi *et al.* 2022; Kou & Gu 2016). To make multimodal travel more attractive to the commuter we should make their first and last mile connectivity easily accessible. Major portion of commuters prefer ride hailing services as their first and last mile mode. But, as the travel demand grows there is more and more demand of such vehicles at any instant. But we cant keep on increasing the supply of such vehicles else the network will become congested and the overall speed of the network will reduce as shown on RMP 2031 study of Bengaluru. So we should think of some other alternative ways. At the same time we should respect the commuters waiting time as well or else again they will move away from using multimodal transport. As our focus is

to bring services to the people not people to the services.

1.3 Contribution

To determine the optimum fleet sizing strategy for a railway station, the first step is to identify the different types of trains accessing that station. These trains can either be starting from the station, terminating at the station, or passing through the station with stops. Next, it is essential to gather information on their timings, coach composition, and, most importantly, passenger loading.

Additionally, understanding passengers' preferences for different modes is crucial. This includes how sensitive they are to factors such as travel time, waiting time, travel cost, travel distance, and transfers. It is also important to consider their individual characteristics, such as gender, age, income group, and education level.

Other key details include whether passengers are traveling alone or with a group (such as family) and whether they are carrying luggage. If they are carrying luggage, it is important to categorize it. To collect this data, a survey should be conducted at the station, either upon arrival or before departure of the trains. The survey will also help identify where passengers are going or where they are coming from.

After collecting the necessary details, the next step is to decide which tool to use and what output is expected based on the inputs. For agent-based modeling, a widely used simulation software is AnyLogic. In AnyLogic, each commuter can be modeled as an agent, and their attributes can be represented through parameters. It also allows the incorporation of GIS maps, enabling the computation of travel time, waiting time, and includes the built-in OptQuest optimization tool, which is based on heuristics.

Using the input data, we can determine the mode share and, consequently, the number of FMLM vehicles needed to serve all commuters at any given moment of the day while keeping their delays within acceptable limits. However, some passengers might choose transit for their FMLM services but may not be able to access it due to excessive waiting times or overcrowding. As a result, these passengers may switch to IPTs, leading to an increase in the number of IPTs in the system. This, however, does not contribute to improving the system.

To address this, we can integrate the bus schedules in a way that ensures more buses are available when high-volume trains are arriving. This would provide better coverage of routes, allowing passengers to easily reach their destinations. By keeping their waiting times within the tolerance limit, we may reduce the number of passengers lost from the buses. This, in turn, could reduce the number of vehicles in the system, as the focus is on moving people, not just vehicles. Additionally, improved service quality and increased coverage may encourage more passengers to switch from private vehicles or IPTs to buses, further reducing the number of vehicles on the road.

Finally, we can analyze the increased average speed of vehicles in these areas to find the "sweet spot," where congestion levels, commuter waiting times, and average network speed are all optimized.

The rest of the report is structured as follows - Chapter 2 discuss about literature review, Chapter 3 shows the research gap and objective of the study, Chapter 4 gives the detailed discussion about the study area and data , Chapter 5 shows the detailed methodology , Chapter 6 shows all the results of base and policy case scenario followed by insights from those results and sensitivity analysis of some important parameters, Chapter 7 discuss about the conclusion and possible future extension of the study and then references to conclude the report.

Chapter 2

Literature Review

2.1 First and last mile connectivity of public transport

The FLM connectivity plays an important role in improving public transit usage. Enhancing FLM connectivity is the key in reducing car usage and improving the appeal of sustainable transport modes, such as walking, cycling, and public transit services (Lu, Kimpton, *et al.* 2024, Gupta *et al.* 2018, M. Meng *et al.* 2016). Lu, Kimpton, *et al.* 2024 has classified the FLM travel modes into four primary categories: conventional sustainable transport modes (such as walking, cycling, and feeder buses), motorized vehicles (including cars, taxis, and car-sharing), emerging transport modes (like autonomous vehicles and micro-mobility options such as e-scooters and bike-sharing), and informal public transport modes (e.g., rickshaws, jitneys, and pedicabs).
Daniels & Mulley 2013 and Givoni & Rietveld 2014 has shown that commuters are more comfortable in walking longer distances to access railway stations that to bus stop. Zuo *et al.* 2020 showed through spatial analysis that better transit accessibility increases public transport usage. P. P. Koh & Wong 2013 and P. Koh & Wong 2015 has shown that commuters avoid walking during hot, humid and rainy weather conditions.
Zgheib *et al.* 2020 has found that households with multiple vehicles are more likely to drive the FLM than households with fewer vehicles. Azimi *et al.* 2021 has shown that commuters are more likely to take taxi to access the public transport stations when they need to carry heavy goods. Givoni & Rietveld 2014 has shown that commuters are willing to drive longer distances to access public transport stations if we improve the service quality of public transport like punctuality, service frequency and cost.

Chen *et al.* 2022 and M. Kim & Cho 2023 has found that micro-mobility helps commuters to access destinations in less time and lower costs as well. In addition to that these are sustainable transport mode as well and can help in reducing vehicular emissions. Yap *et al.* 2016 and Abe 2021 have shown that with the increase in income level commuters preference towards autonomous vehicles as FLM mode increases. Huang *et al.* 2021 has found that while accessing the public transport using autonomous vehicles, the requirement of parking reduces.

Chidambara 2016 and Saffan & Rizki 2018 have mentioned that in the developing nations due to the lack of coverage, safety and affordability of feeder bus services and also not owning a vehicle leads to the usage of informal public transport modes and these are faster, cheaper and can serve area with narrow roads (Fillone & Mateo-Babiano 2018 & Kumar & Roy 2019). Saffan & Rizki 2018 and Fillone & Mateo-Babiano 2018 have also shown that people switch to informal mode of transport due to poor walking environment and narrow roads. Kumar & Roy 2019 has mentioned, in India (specially in Delhi) to solve FLM problem we can shift to E-rickshaw from auto-rickshaw as it does not lead to air pollution.

2.2 Agent based simulation modeling on public transport

Agent-Based Modeling (ABM) plays a critical role in simulating urban mobility systems, particularly in public transport, by modeling the interactions between agents (such as passengers and vehicles) and the environment. These simulations help evaluate the impact of various policies and infrastructure changes, providing valuable insights for optimizing traffic flow, reducing congestion, and improving service efficiency. By integrating different modes of transport, ABM can offer solutions to challenges like first and last-mile connectivity and demand-responsive transport systems (Stevens *et al.* 2022,S. Wang *et al.* 2022, Franco *et al.* 2020, Divasson-J *et al.* 2025). Franco *et al.* 2020 demonstrated that utilizing an Agent-Based Model (ABM) integrated with Demand Responsive Transit (DRT) services can reduce private car usage, leading to a 30% increase in public transport patronage. Using an Agent-Based Model (ABM), S. Wang *et al.* 2022 analyzed the competition among multiple Automated Mobility-on-Demand (AMoD) operators and how their operational strategies impacted traffic performance. They found that optimizing vehicle assignment algorithms reduced average waiting times by up to 24%, while increasing fleet sizes boosted demand but led to higher congestion, thus worsening travel times and service quality for all operators. Stevens *et al.* 2022 found that combining dynamic ride-pooling with fast-charging strategies led to a 6% reduction in fleet size while optimizing vehicle usage, resulting in a 49% increase in profit.

Badland *et al.* 2013 has developed a simple ABM to enhance neighborhood walkability and evaluate the accessibility of key services. By using this tool, stakeholders can test various built environment scenarios to optimize first and last mile connectivity, improving pedestrian access to services like public transport nodes and schools. Inturri *et al.* 2019 shows that through ABM while integrating flexible transit systems (such as Demand Responsive Shared Transport (DSRT)) can significantly improve first and last mile connectivity and in the case study he showed with 6 vehicles operating, 78% of the maximum demand was satisfied, and travel distances were reduced by 26% .

2.3 Studies on railway station

A railway journey is almost always part of a journey 'chain' that includes a journey to, and later from, the railway station by different modes of transport. The integration of these components is essential to achieve a continuous travel, door-to-door when using the rail (Givoni & Rietveld 2007). Wardman & Tyler 2000 discovered that people are much less likely to take the train if they live far from the station. Their findings suggest that making it easier and faster to get to the station is a key way to encourage more people to travel by rail. Martens 2004 by improving the access to railway stations by public transport and non-motorized modes can lead to car free life style. Givoni & Rietveld 2007 showed that passengers will accept longer journey time and distance for the access journey than for the egress journey to the railway station. Givoni & Rietveld 2007 also showed the commuters sensitivity to travel distance in FLM mode choice to railway station like for short distance its mainly NMT but as distance increases they shift to public transport. Givoni & Rietveld 2007 also showed that bicycle is an important mode at the home end station but less at the activity end station. Martens 2004 and Givoni & Rietveld 2007 have shown that car ownership does not play an important role in the choice of access mode to the station. Stam *et al.* 2021 shows that for home end trips commuters mostly preferred buses (45%) and bicycle (41%) while for the activity end trips walking is preferred (67%) most frequently over other modes from a railway station.

Some of the significant research papers are briefed in more details in the tables 2.1, 2.4, 2.3, 2.5 & 2.2 based on FLM fleet sizing, FLM mode choice from railway station, Pedestrian Simulation using AnyLogic etc..

Table 2.1: Some important literature in the context of FLM fleet sizing

Sl no.	References	Study Area	Objective	Methodology	Tools / Models used	Key findings
1	Bhatnagar *et al.* 2024	Metro transit network of Bengaluru, India	• To improve first- and last-mile (FLM) connectivity to metro systems by optimizing FLM fleet allocation and operations. • Minimize unserved passenger requests (lost demand) • Quantify benefits of ride-sharing for FLM services in reducing waiting time, congestion, and operational costs.	The authors use a hybrid approach combining agent-based simulation and optimization to model FLM vehicle allocation. The simulation replicates passenger and vehicle behavior with statecharts, while the optimization model allocates vehicles using a knapsack-like formulation. Four scenarios are simulated based on vehicle sharing condition.	AnyLogic, CPLEX, Google OR-Tools, QGIS	• ILP-based allocation reduced lost demand by up to 81% compared to baseline. • Joint ride-sharing decreased vehicle kilometers traveled (VKT) by 27%, reducing congestion and emissions. • Distance-based pricing generated higher revenue per vehicle than flat fares. • Identified peak periods (7–8 PM, 9–10 AM) with highest unserved demand.

2	X. Guo & Zhao 2024	Chicago, USA	The primary goal is to minimize passenger disutility (waiting and walking times) by optimizing transit network design, AMoD fleet sizing and allocation, and multimodal pricing and jointly.	The researchers propose TCMUM-AMoD, an optimization framework using a Mixed Integer Non-Linear Program (MINLP) to optimize rail/bus frequencies, AMoD fleet allocation, and pricing by considering commuters mode and route choices using a Nested Logit (NL) model.	Gurobi 10.0.2, Python programming language	• All-bus scenario produced the lowest passenger disutility at 80% downtown demand. • Adding AMoD reduces walking and expected waiting time • Optimal balance : AMoD best for local FLM trips & Transit is best for long distance commuters
3	Lokhandwala & Cai 2018	New York City (NYC), USA	To quantify the environmental and energy benefits of dynamic ride sharing using traditional and autonomous taxis through agent-based modeling.	Used agent-based modeling (ABM) to analyze the potential impacts of dynamic ride-sharing using both traditional taxis (with shifts) and shared autonomous taxis by considering capacity constraint, individual heterogeneity & varying participation level	AnyLogic	• Ride sharing increased occupancy rate (from 1.2 to 3) and decreased total travel distance upto 55%. • Ride sharing through autonomous vehicles can reduce the fleet size upto 59%. • Daily CO_2 emission is lowered by 725 metric tonnes.

4	Shen *et al.* 2018	Singapore	To assess how integrating shared Autonomous Vehicles (AVs) with existing public transit systems can improve the first-mile service and overall system performance, while maintaining or enhancing service quality and financial sustainability.	They have used agent-based modeling for simulating the integration of AVs with the existing transit system.	AnyLogic	• The average Out-of-vehicle time (OVT) for passengers in the integrated AV-PT system is improved compared to the bus-only system. • Ridesharing with AVs reduces OVT. • The integrated AV-PT system is financially sustainable within the tested fleet size range and pricing model.

Table 2.2: Some important literature in the context of FLM mode choice to or from railway station

Sl no.	References	Study Area	Objective	Methodology	Tools / Models used	Key findings
1	Givoni & Rietveld 2007	Netherlands.	To analyze the modes of transport used by the commuters in accessing or egressing to or from railway station.	They have used passenger satisfaction survey data, conducted by Dutch Railways. Afterwards descriptive analysis was performed to study access and egress mode choice to railway stations, followed by linear regression models to assess how station quality and access facilities influence overall satisfaction with rail travel.	Statistical data analysis & linear regression	• Car ownership does not automatically translate into car-based access. • Walking, cycling, and public transport dominate access to and egress from railway stations.

2	Araghi *et al.* 2022	Delft Campus railway station, Netherlands.	To examine how environmental (weather) and situational (luggage) contexts influence travellers' preferences for emerging first/last-mile modes at a small multimodal hub.	A stated preference survey was conducted with respondents at the Delft Campus station. The survey included a choice experiment with five emerging transport modes and context scenarios to analyze their influence on passengers' mode choice.	Multinomial Logit Model (MNL) & NL	• During rainy days people carrying luggage has lower preference towards shared bike/scooters • During adverse conditions commuters are having higher preference for automated vehicles(AVs).
3	Kou & Gu 2016	Shanghai railway station, China	To analyze how passengers choose access modes to major railway stations in a metropolis considering luggage type and Per-Capita Family Income (PCFI)	A MNL model was used to estimate the access modal split, with data collected from a survey at the Shanghai railway station. The survey included factors such as access time, fare, luggage types, and per capita family income.	MNL	• Passengers with higher income and light luggage prefer taxi. • Low income people with heavier luggage prefer bus or metro. • High income people with heavy luggage would give little preference to Bus or metro.

4	Stam *et al.* 2021	Almere Centrum railway station, Netherlands	To provide insights into travelers' preferences for existing and emerging means of first/last mile (FLM) transport at railway station.	A mode choice experiment is conducted with travellers at Almere Centrum, using four distinct future scenarios based on varying levels of shared vehicle and on-demand ride service availability.	Google forms and Sankey diagram (to visualize mode shift)	• In no shared or on-demand services scenario → travellers rely on private vehicles and buses. • In shared vehicles replacing private cars scenario → buses gain mode share (31% to 50%) • On availability of both shared vehicles and on-demand services → buses and two wheelers usage shows a decline.
5	Lieu & Akar 2025	Atlanta, USA	To understand the mode choice behaviors of rail users for FLM travel based on demography and various influencing factors.	They have used survey data from rail users and street images from Google Street View to understand how people choose modes for FLM trips. Then through MNL they have examined how personal characteristics, transit service quality, and street design influence these choices.	MNL, OSM, Google street view etc.	• Bus usage for FLM trips is directly proportional to its service quality. • Street-level built environment features such as greenery significantly encourage walking. • Parking availability at rail stations is not a significant factor for influencing FLM mode choice.

Table 2.3: Some important literature in the context of FLM solution through matching algorithm

Sl no.	References	Study Area	Objective	Methodology	Tools / Software used	Key findings
1	Kumar & Roy 2019	Twin Cities, MN, USA	To integrate peer-to-peer ridesharing with schedule-based transit to provide an efficient solution to the FMLM problem, in order to improve access to transit stations and encouraging transit use.	Developed an algorithm for transit passenger routing in a schedule-based transit network and a matching optimization program to match riders and drivers up to the first mile of the transit trip.	Python 2	• Integration of ridesharing with transit can reduce the required fleet size to serve the same demand. • Matching efficiency improves with higher time flexibility for both riders and drivers. • Walking and waiting time of the rider reduces due to efficient matching.

Table 2.4: Some important literature in the context of FLM mode choice under different setting

Sl no.	References	Study Area	Objective	Methodology	Tools / Models used	Key findings
1	Lu, Prato, *et al.* 2022	Brisbane, Australia	The goal is to explore how intra-household interactions influence travel mode choices for FLM travel, considering joint travel decisions and household characteristics like car ownership and the presence of children.	A latent class nested logit model is used to capture intra-household interactions by segmenting households based on their modality style. In which - • Upper level captures → decisions of joint or individual travel • Lower level captures → mode choice preference for each type of FLM travel	Latent class NL model	• Joint travel was predominantly car based (78% mandatory, 67% discretionary) • Individual travel showed higher walking shares (57-64%) • Households with young children and high car ownership → more likely to travel jointly by car • Smaller households with fewer young children → prefer individual or walking-based FLM travel

Table 2.5: Some important literature in the context of pedestrian simulation through AnyLogic

Sl no.	References	Study Area	Objective	Methodology	Tools used	Key findings
1	A. Verma *et al.* 2020	Kempegowda International Airport (KIA), Bangalore	To develop a simulation model that - • Simulates the domestic passenger flow at the airport. • Predicts the impact of various policy scenarios on system throughput. • Helps airport authorities improve the efficiency of terminal operations and passenger service time.	They have used an agent-based simulation model for the Kempegowda International Airport (KIA), Bangalore to understand the impact of various proposed policy scenarios focused towards increasing the throughput (percentage of arrived passengers served in a given time) of the airport terminal.	AnyLogic, BIO-GEME	• Introduction of self-check-in kiosks (scenarios 5 and 6) can significantly improved system throughput. • Female passengers tended to spend more time in the system compared to male passengers. • Online check-in and self-check-in kiosks significantly reduced service time.

Chapter 3

Research Gap and Objective

3.1 Research gap

There have been many studies related to FLM connectivity of public transport, particularly focusing on mode choice at railway stations. These studies have considered individual attributes such as gender, age, income, and sensitivity to travel time, travel distance, waiting time, travel cost, etc. They also consider system-level attributes, such as the quality of road infrastructure and the service quality of public transport. Additionally, researchers have examined the drawbacks of ride-hailing services, such as low occupancy levels and increased empty miles traveled. To address these issues, concepts like ride-sharing and AMoD (Autonomous Mobility-on-Demand) have been proposed to eliminate parking facility needs and the shifting problems associated with traditional ride-hailing and taxi services.

Furthermore, some studies have utilized agent-based simulation models to analyze changes in FLM fleet size, travel distance, and CO_2 emissions under different scenarios, such as Business-As-Usual (BAU), replacing non-shared rides with shared rides, and integrating AVs with public transit.

Most of these studies have either focused on larger networks or, at the station level, mainly on mode choice for the FLM leg. However, what happens when over 1,000 passengers suddenly deboard from a train? How will they choose their modes based on factors like luggage, group or individual travel, and other attributes? Additionally, what would be the required FLM vehicle fleet size to keep commuters' delays within specified limits? How can we improve public transport usage to reduce the FLM fleet size?

While some relevant studies exist for metro networks and passengers coming out of airport terminals, the situation at large railway station hubs remains largely unexplored. Although researchers have addressed these issues in bits and pieces, often across different parts of the globe, this presents an opportunity to explore a new dimension of FLM connectivity from a railway station perspective.

3.2 Objective of the study

This study aims to develop a simulation-based framework for evaluating fleet sizing strategies at a multimodal interchange, using agent-based modeling (ABM) to capture the heterogeneous mode choice behavior of commuters. Commuter preferences are modeled as a function of individual attributes, including sensitivity to travel time, travel cost, and waiting time, as well as time of travel, travel mode (individual or group), luggage weight, and income category.

The framework is applied first under a Business-as-Usual (BAU) scenario to establish baseline fleet performance. Subsequently, a Schedule Integration Scenario is introduced , wherein BMTC bus availability is dynamically increased in response to high-volume train arrivals at the interchange, simulating a demand-responsive operational strategy. The resulting changes in fleet sizing recommendations and system performance are then assessed comparatively.

The proposed framework is validated against real-world conditions at the Yeshwantpur Multimodal Interchange, serving as a case study to test the model's applicability and transferability to operational transit planning contexts.

Chapter 4

Study area and data

4.1 Study area

For this study, to test the performance of our model we are considering Yeshwantpur Interchange area i.e. Yeshwantpur Indian Railway Station area whose latitude and longitude read as follows 13.02°N and 77.55°E. This railway station is located at the North Western part of Bengaluru which is one of the area that lies within the heart of city. Apart from railway station there is BMRCL metro station, BMTC bus stops and upcoming corridor 1 and corridor 2 of sub-urban. And for our analysis we will consider a buffer of radius 5 km around that interchange as shown in the figure 4.1. For that we have computed the population within this buffer as 752843 from the TIF file downloaded from WorldPop Research Group 2020 and then we have created 1000m $\times$ 1000m square grid within the study area and population inside each grid is shown in the figure 4.2.

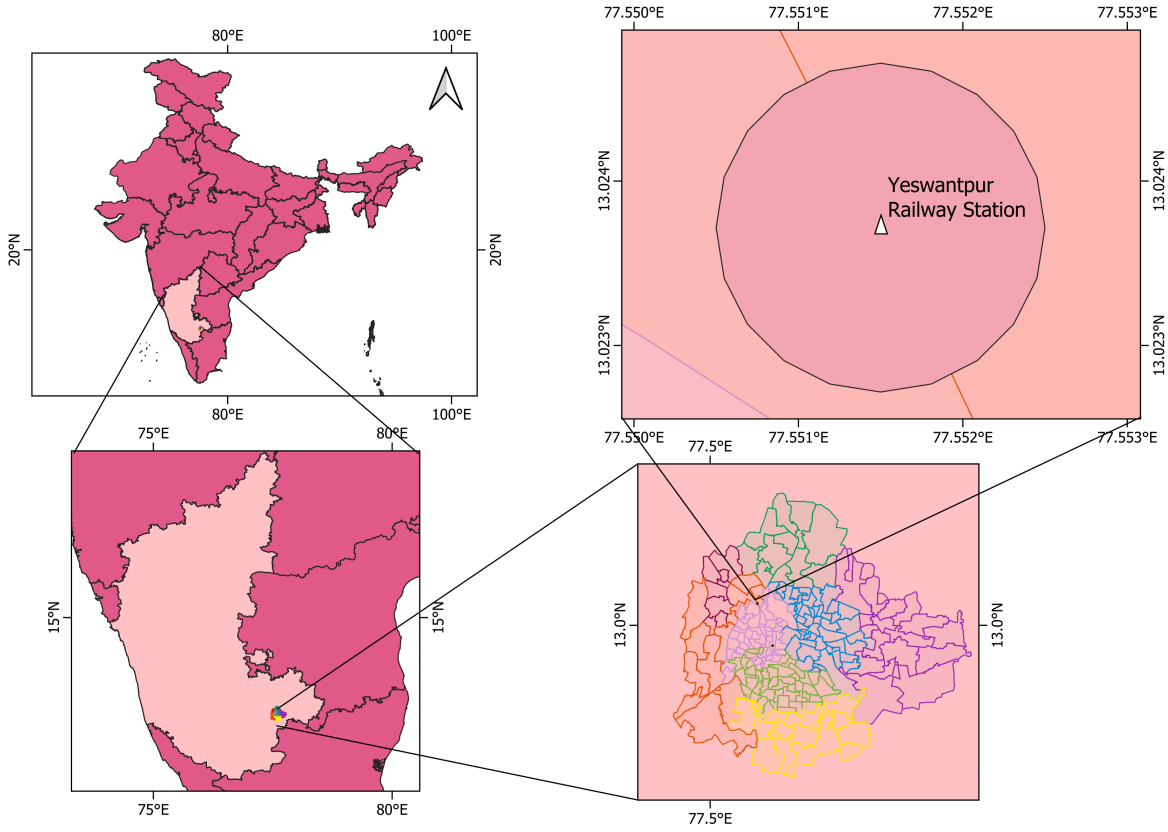


Figure 4.1: Study area under consideration

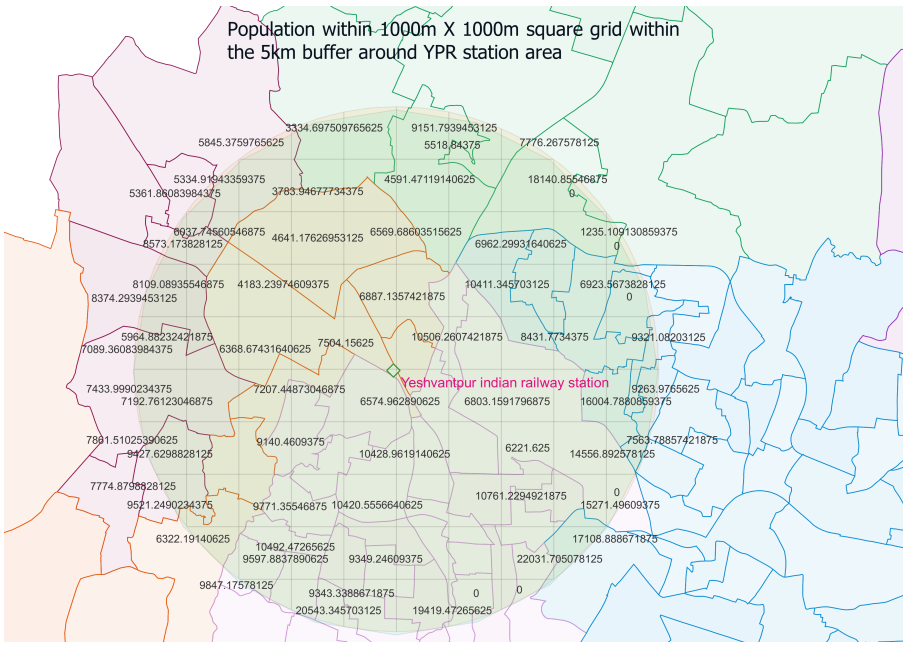


Figure 4.2: Population within each grid

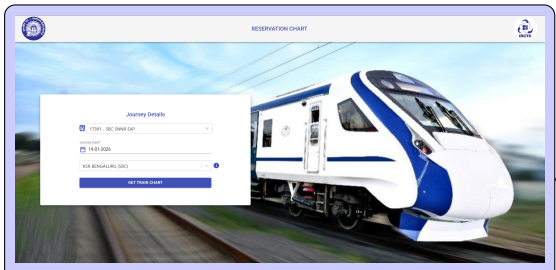
4.2 Data collection

For this study we need the following data

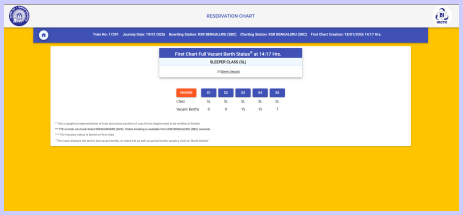
- Train schedule and delay data at YPR station (Google LLC 2026).
- Boarding and deboarding passenger counts at the interchange (IRCTC reservation charts 2025).
- BMTC GTFS data for bus availability
- BMRCL Green Line shapefile and schedule (Bangalore Metro Rail Corporation Limited (BMRCL) 2025).

4.2.1 Daily ridership data collection at Yeshwantpur Railway station

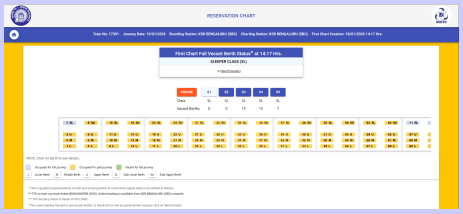
To collect daily ridership data i.e. nos of passengers boarding and deboarding from different times at different platforms are collected through or to be honest scrapped through web scrapping tool sync playwright (Microsoft Corporation 2024) from the indian railways official website for reservation charts section (IRCTC reservation charts 2025). Indian railway provides all reservation coaches berth detailing like for that berth from where to where it is occupied and vacant etc. The detailed process is explained in the flowchart 4.3. However, IRCTC does not not provide nos of passengers are boarding from YPR or nos of passengers are deboarding at YPR from the general coach. But from our observation it has been seen that for express trains all of the seats get occupied at the YPR platform itself for trains starting at YPR and similarly for trains terminating at YPR also almost having nos of passengers deboarding from general coach equal to their seating capacity. So, we will assume nos of seats in a general coach and nos of coaches on that trains is the passengers demand for general coach and will add that value to the values obtained from reserved coach. for trains passes through YPR we will assume the similar proportion to that of reservation coach demand.



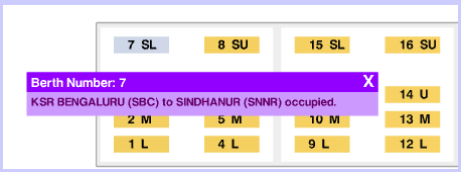
Load IRCTC reservation chart website and enter required details (IRCTC reservation charts 2025)



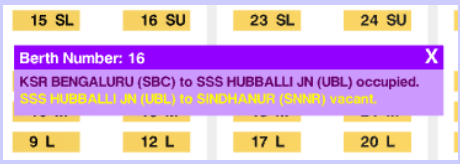
Load the coach composition page



Enter into a particular coach to get the berth details



Light violet colored berths indicates that berth is occupied for full journey



Yellow colored berth indicates it is occupied for part journey and collect that berth details

Coach no	Berth no	Occupied upto YPR	Occupied from YPR	Full route details
S1	1	NA	NA	KSR to SNNR occupied
S1	2	NA	NA	Vacant for full journey
S1	3	Yes	Yes	KSR to YPR occupied YPR SNNR occupied
S1	4	NA	Yes	KSR to YPR Vacant YPR SNNR occupied

Tabulate those details in the following form for every train

Train Number	Passengers Deboarding at YPR	Passengers Boarding at YPR
11311	312	0
11312	14	236
12079	0	332
12080	183	0
12089	0	153
12090	342	0

Tabulate those details in the following form for every train

Figure 4.3: Entire data collection (both scrapping and analysis) mechanism

4.2.2 Data collection results

Before getting into the daily boarding, deboarding values of the commuters we need to identify the nos of trains are coming to or going from or passing through YPR station. For that we took the help of MakeMyTrip 2026 website where they have mentioned clearly as separate sections like trains are terminating at YPR, trains are starting from YPR and trains are passing through YPR. But, later we found that IRCTC has discontinued some of the trains and inserted some new and for that cross verification and updated list we tally with the Google LLC 2026 for each OD pair got from previous list.

Total nos of trains accessing YPR platform : As,we have discussed there are mainly three types of trains i.e. some are starting at YPR, some are terminating at YPR and some of them are passing through YPR on daily basis and their detailed value you can read from the figure 4.2.2. Also these trains are coming from or going to different parts of the country as you can see from the figure .

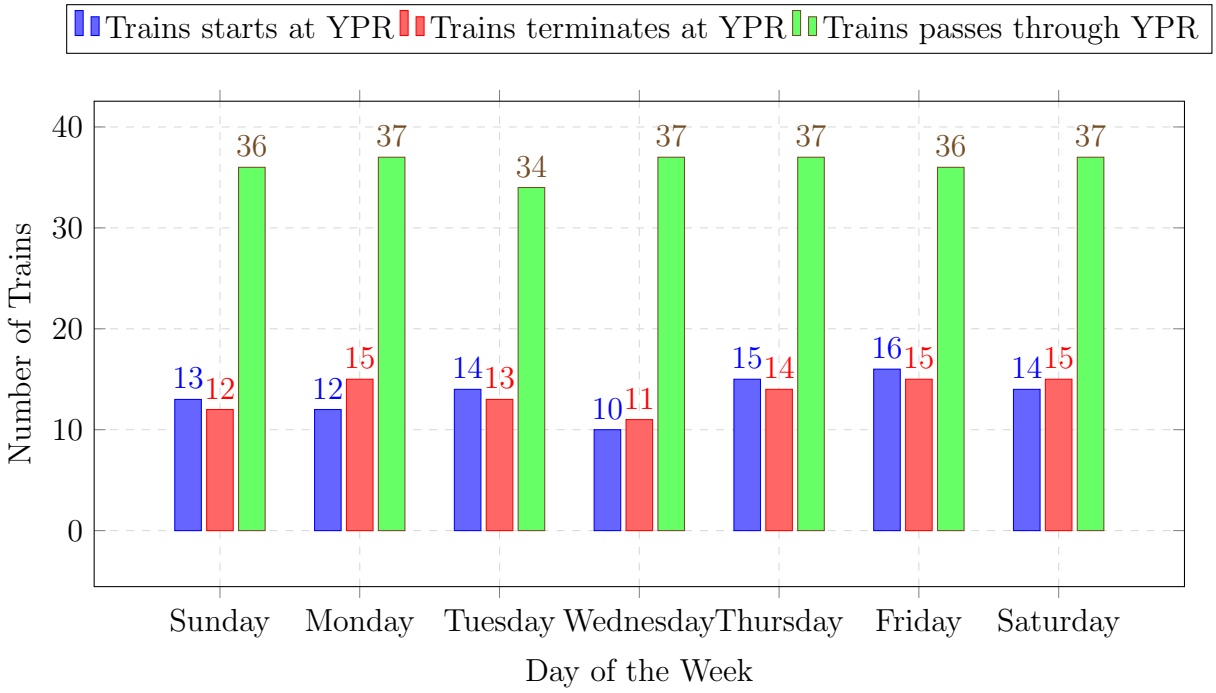


Figure 4.4: Day-wise distribution of trains at YPR station

Temporal variation of daily boarding and de-boarding of passengers at YPR station : Using the methodology we have discussed in the figure 4.3, we have extracted data for one and half months and that too for the reservation coaches. However, for this study we will be using weekday (one of five) and weekend (one of two) trends. From the following figure (4.5 and 4.6) you can see the average weekday and weekend trend for the same. However, the exact data that is fed in the model for the case study is slightly different from these. For that we have considered some provisions for the general coaches passengers also which is discussed in the following section.

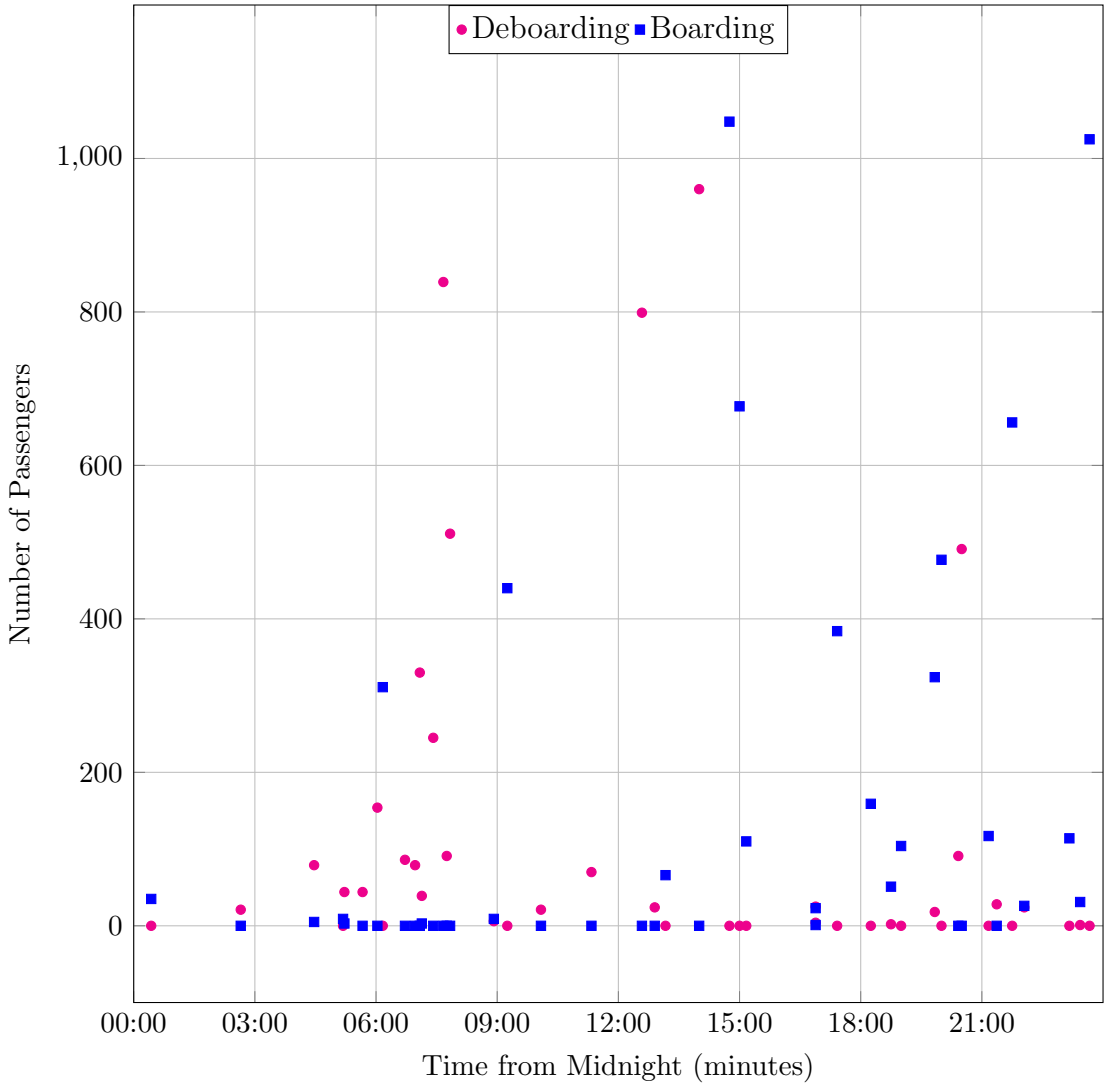


Figure 4.5: Average nos of passengers boarded and de-boarded at YPR station on Weekday

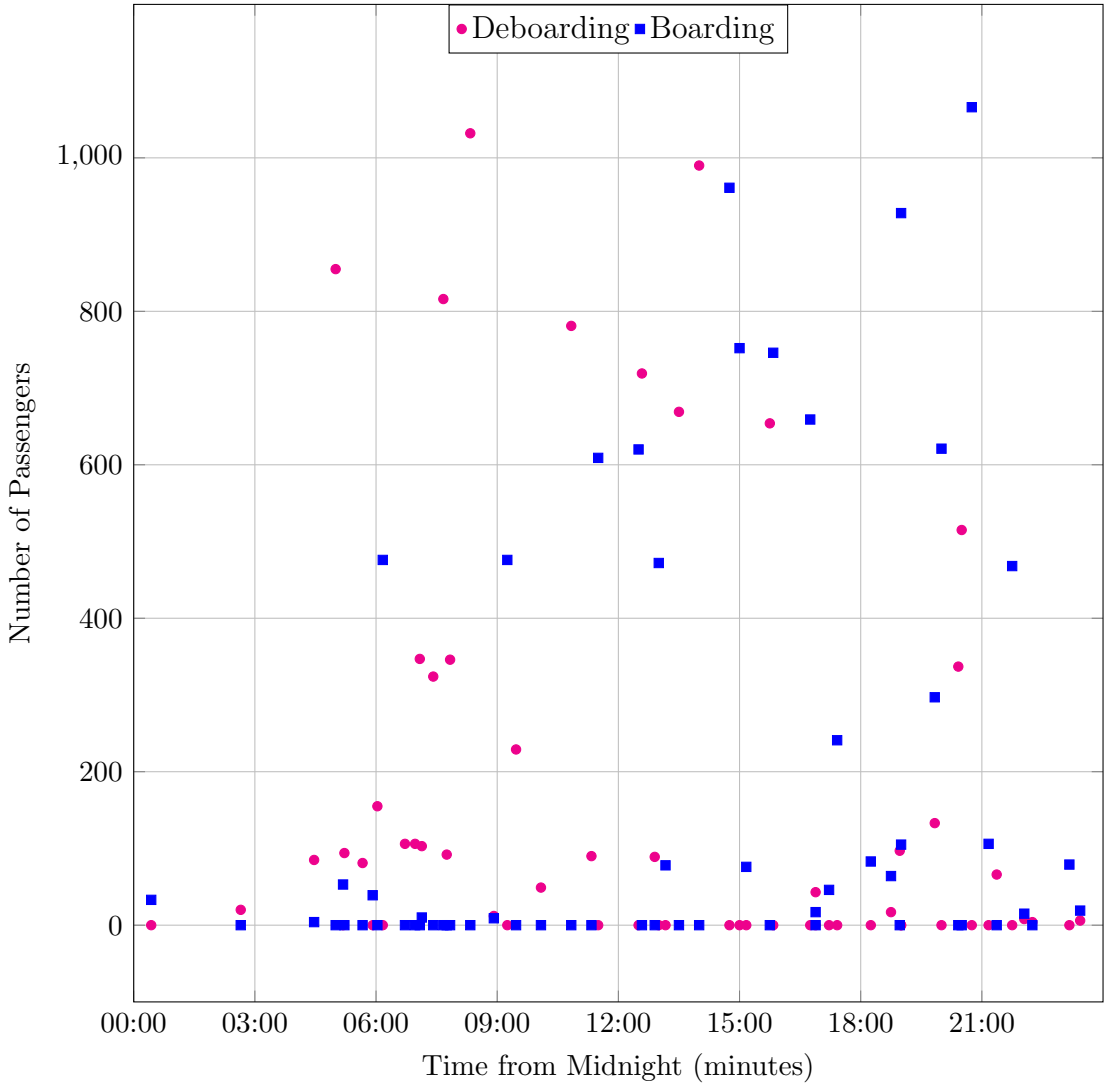


Figure 4.6: Average nos of passengers boarded and de-boarded at YPR station on Weekend

Daily variation of passengers accessing the YPR railway station : In the above figures we have shown the temporal and daily variation of passengers accessing the YPR station area. Though it is important to have an estimate that how many passengers are accessing the station for boarding and de-boarding on daily basis.

We have collected data for six weeks from the beginning of the January 2026 to the mid of February 2026 and in the following figure 4.7 we have shown the average values of those . Though the numbers mentioned there are only for passengers accessing the station area through reservation coaches only. And from our figure 4.3 we can see on an average daily 27 express trains either starts or ends at YPR station and each of them having 4 general coaches (mostly) and if we consider 84 seating passengers per coach then additional 336 passengers per train and almost additional 9000 passengers per day that to through those 27 starting or terminating trains only. And if we consider additional 50 passengers per day (both boarding and deboarding) from each trains passing through the YPR from their general coaches (most of them are also having 4 general coaches) and we having at least 36 such trains per day (refer fig 4.3). Then that nos become additional at least 1800 passengers per day. Hence, by considering these two criteria we can say another additional 11000 passengers per day (refer 4.1 & 4.2) who are accessing the station from general coaches of all the trains accessing YPR railway station.

Now, from the plot 4.7 and our discussion above we can comment at least 25000 passengers are accessing the YPR railway station on any given day and that number can be as high as near to 33000 or maybe more than that.

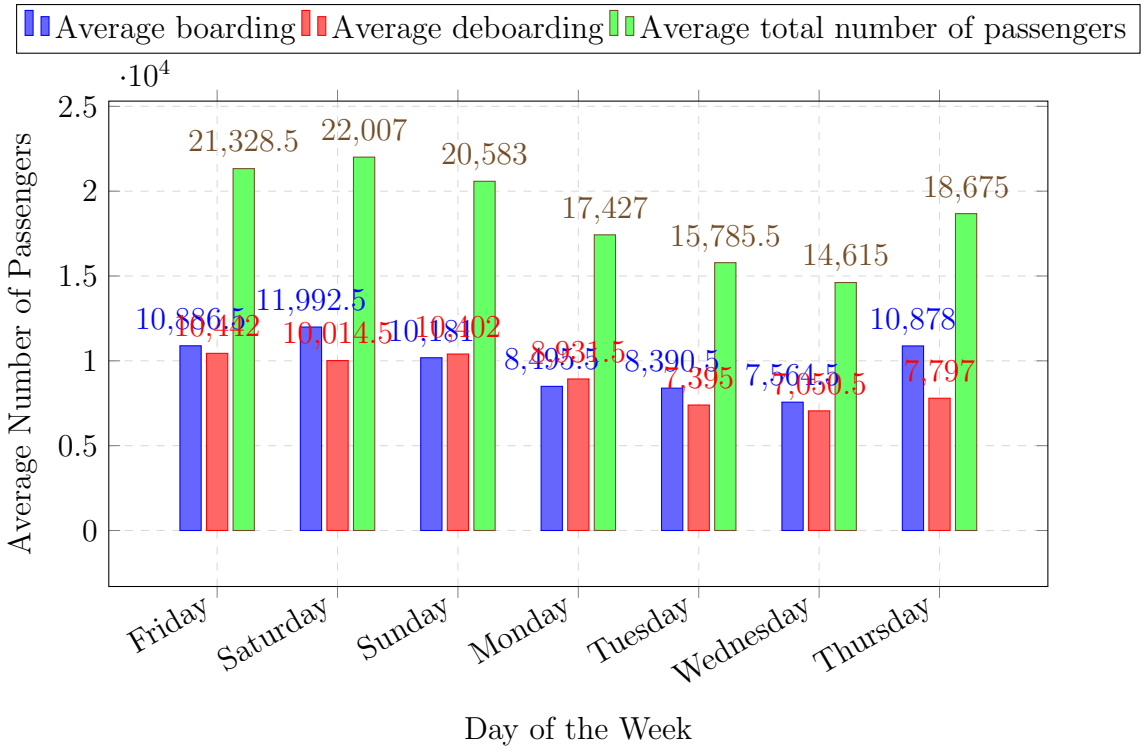


Figure 4.7: Day-wise average boarding, deboarding, and total passenger volumes

Apart from the above mentioned facts and statistics (shown in in the figure 4.7)

there is something more to consider. As, while Indians are out for a journey through Railway they do not access the station alone. Along with them most of the times their near and dear ones will come to see them off also same goes for debaording to pick them up. That adds on some amount of additional passengers load and consequently requirement of more FMLM services. In addition to that passengers accessing YPR specially the trains passing through YPR, they will order food online and which is delivered at the station during the dwelling of the trains at the platforms. Those delivery partners mostly use two wheelers to access the station and that also add some two wheelers demand on the network.

Considering general coaches passengers demand : It is impossible to get the exact estimate of the general coach passengers without the actual ground data. But, for our study to taken them also into the account, we have adopted following methods to add on some passengers in the form of general coach passengers to the existing reservation coach passenger. For this we have classified the trains into three categories which are as follows -

- **Weekly long distance train :** Trains which are running upto thrice in a week and traveling more than 24 hours.
- **Weekly overnight train :** Train which are running upto thrice in a week but completing their journey within 24 hours (mostly 12-16 hrs).
- **Daily overnight train :** Trains which are running daily and finishing their journey within 24 hrs.

Now, we know railway general coaches have seating capacity of 84 people. We also have train details data and their coach composition. Now we have added general coach passengers in the following way -

Table 4.1: General coach passengers estimation for different types of trains

Train type	General coach passenger computation
Weekly long distance	1.5 × seating capacity per coach (84) × Nos of coaches in that train
Weekly overnight	1.0 × seating capacity per coach (84) × Nos of coaches in that train
Daily overnight	0.5 × seating capacity per coach (84) × Nos of coaches in that train

By applying this method we add additional 10000 riders on existing daily ridership data which is collected from the reservation coaches and the updated average nos of daily ridership you can see from the table 4.2.

Table 4.2: Final average daily ridership data

Day	Avg Passengers
Sunday	28095
Monday	25080
Tuesday	23982
Wednesday	20335
Thursday	25870
Friday	29050
Saturday	30989

Train delay data

Delay is a most common phenomena in Indian railway. One of the major objective of our study is to take into account the delay arrival of the trains which leads to delayed arrival of the last mile commuter. for that we have collected one month delay data from Google LLC 2026 for all the trains currently accessing the Yeshwantpur railway station area and their details is shown both for weekdays and the weekends in the figure 4.2.2 and 4.2.2 respectively.

In the figure 4.8 we have attached the Google LLC 2026 train interface and you can see the scheduled time of train arrival and departure station wise also below that in red color, the actual time when the train has arrived or departed from that particular station. So, from those two time stamps we can easily get the delay value of that train for that particular train at station of interest. In this study, since we are taking Yeshwantpur interchange as our case study, we have collected all the delay value for the time stamp corresponds to Yeshwantpur station only.

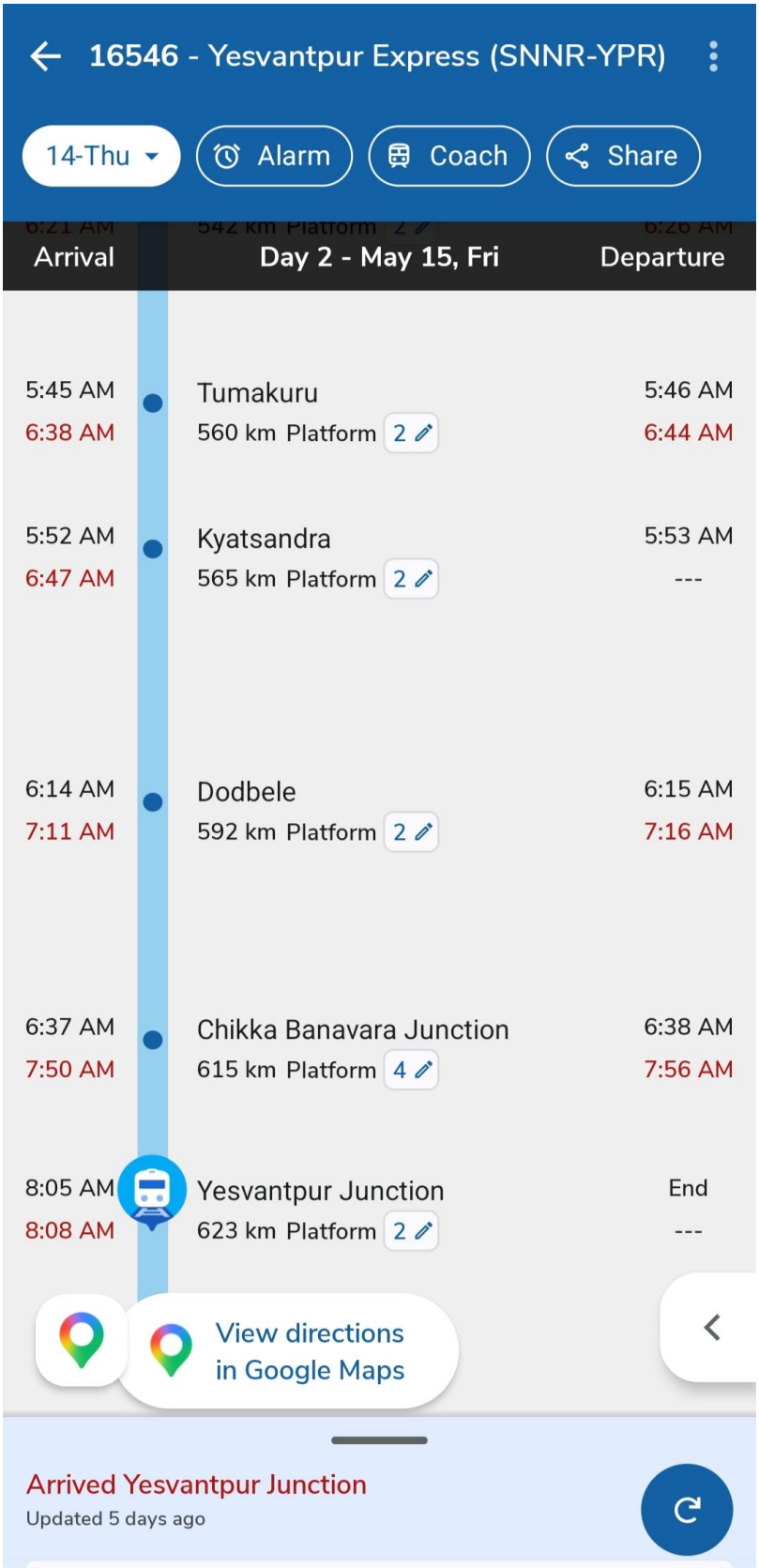
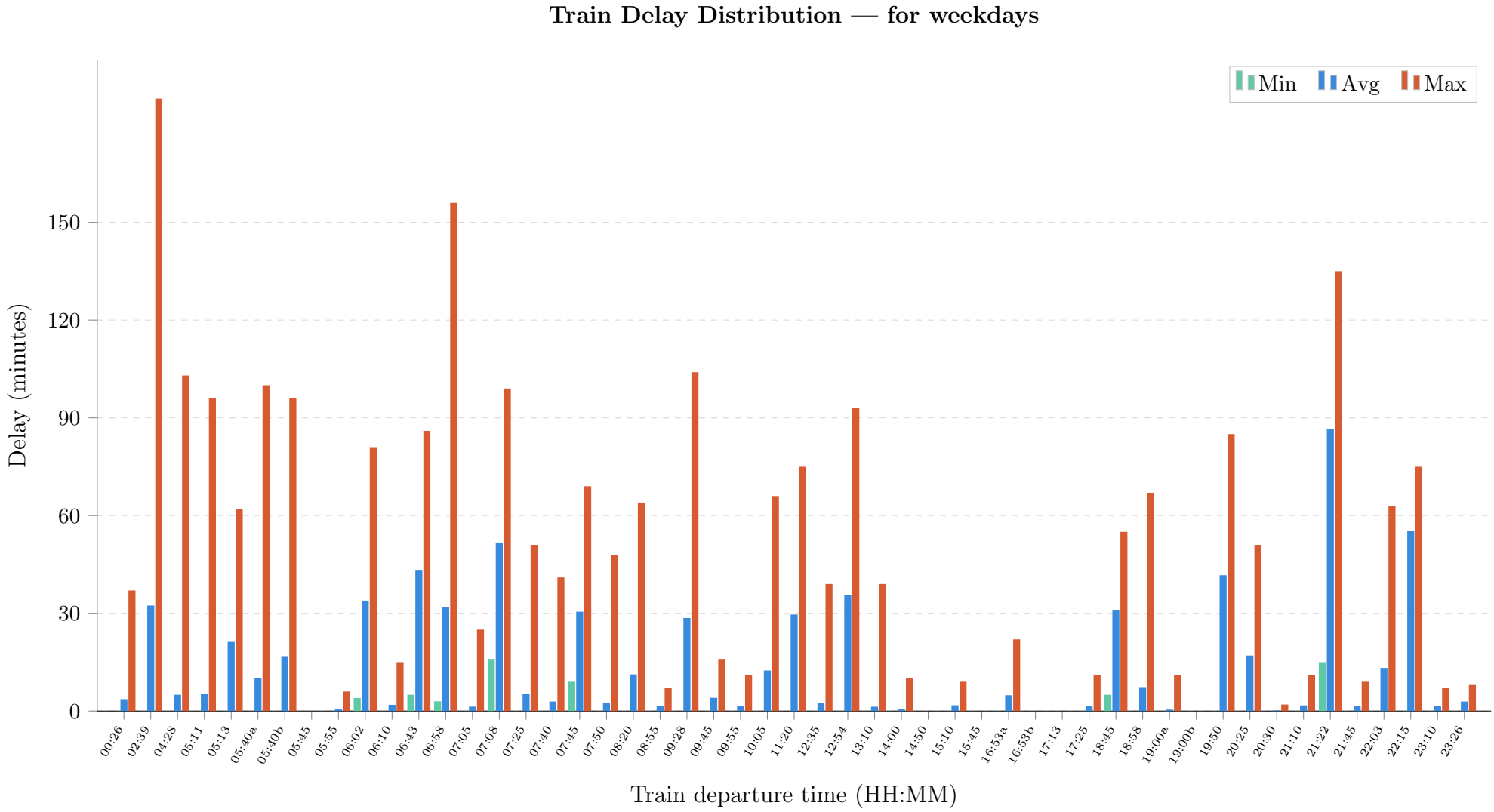
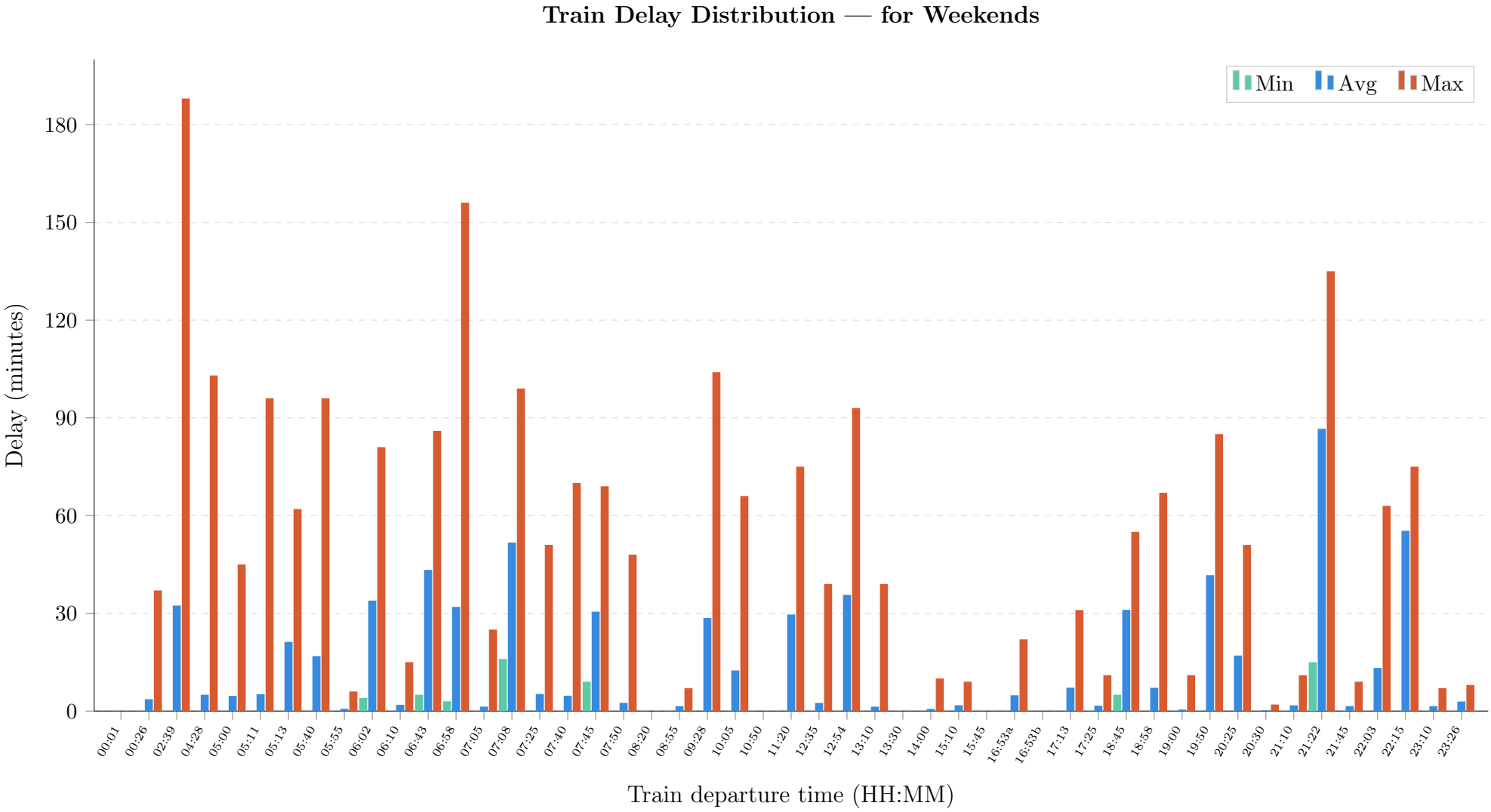


Figure 4.8: Google LLC 2026 interface to get the train delays


Train Delay Distribution — for weekdays
Min
Avg
Max
Delay (minutes)
0
30
60
90
120
150
00:26
02:39
04:28
05:11
05:13
05:40a
05:40b
05:45
05:55
06:02
06:10
06:43
06:58
07:05
07:08
07:25
07:40
07:45
07:50
08:20
08:55
09:28
09:45
09:55
10:05
11:20
12:35
12:54
13:10
14:00
14:50
15:10
15:45
16:53a
16:53b
17:13
17:25
18:45
18:58
19:00a
19:00b
19:50
20:25
20:30
21:10
21:22
21:45
22:03
22:15
23:10
23:26
Train departure time (HH:MM)


Train Delay Distribution — for Weekends
Min
Avg
Max
Delay (minutes)
0
30
60
90
120
150
180
00:01
00:26
02:39
04:28
05:00
05:11
05:13
05:40
05:55
06:02
06:10
06:43
06:58
07:05
07:08
07:25
07:40
07:45
07:50
08:20
08:55
09:28
10:05
10:50
11:20
12:35
12:54
13:10
13:30
14:00
15:10
15:45
16:53a
16:53b
17:13
17:25
18:45
18:58
19:00
19:50
20:25
20:30
21:10
21:22
21:45
22:03
22:15
23:10
23:26
Train departure time (HH:MM)

4.2.3 BMTC GTFS data visualization and catchment creation for both BMTC and BMRCL stops

We have GTFS dataset of BMTC around YPR station and we know there are majorly 7 bus stops (refer 4.3) around the YPR station area through which people can access their FMLM journey and those stops are shown in the figure 4.9.

Table 4.3: Connecting bus stops name and id around interchange area

Connecting Bus Stop Name	Stop Id
YPR New Railway Station (RMC Yard)	21131
YPR RMC Yard	21132
YPR Market	23334
YPR New Railway (Market Side)	27453
YPR Railway Station	27535
YPR TTMC 1	21288
YPR TTMC 2	24016

We will only consider a buffer of 5km around the YPR station area and all the bus routes connecting those YPR interchange hub bus stops. From the figure 4.10 we can clearly see that still there is significant nos of bus stops which is not directly accessible through the routes created by connected routes of those 7 bus stops. In the figure 4.10 all the bus stops shown by green dot tells about all the available bus stops within the 5km buffer around the YPR station area and black dot bus stops indicates the bus stops that are directly accessible through the connected routes (shown in blue mark) of those four bus stops around the interchange area.

Further, we want to create a catchment of 500m around each bus stop within the 500m buffer and coming on that connected routes of those 4 bus stops along with that the metro stations also. Within the 5km buffer around YPR station area mainly green line of BMRCL passes and the combined catchment of BMRCL stations and those bus stops are shown in the figure 4.11. In this figure we have only shown 500m buffer around connected bus stops on the routes passing through the 4 bus stops of interchange area and the Bangalore Metro Rail Corporation Limited (BMRCL) metro stations of green line.

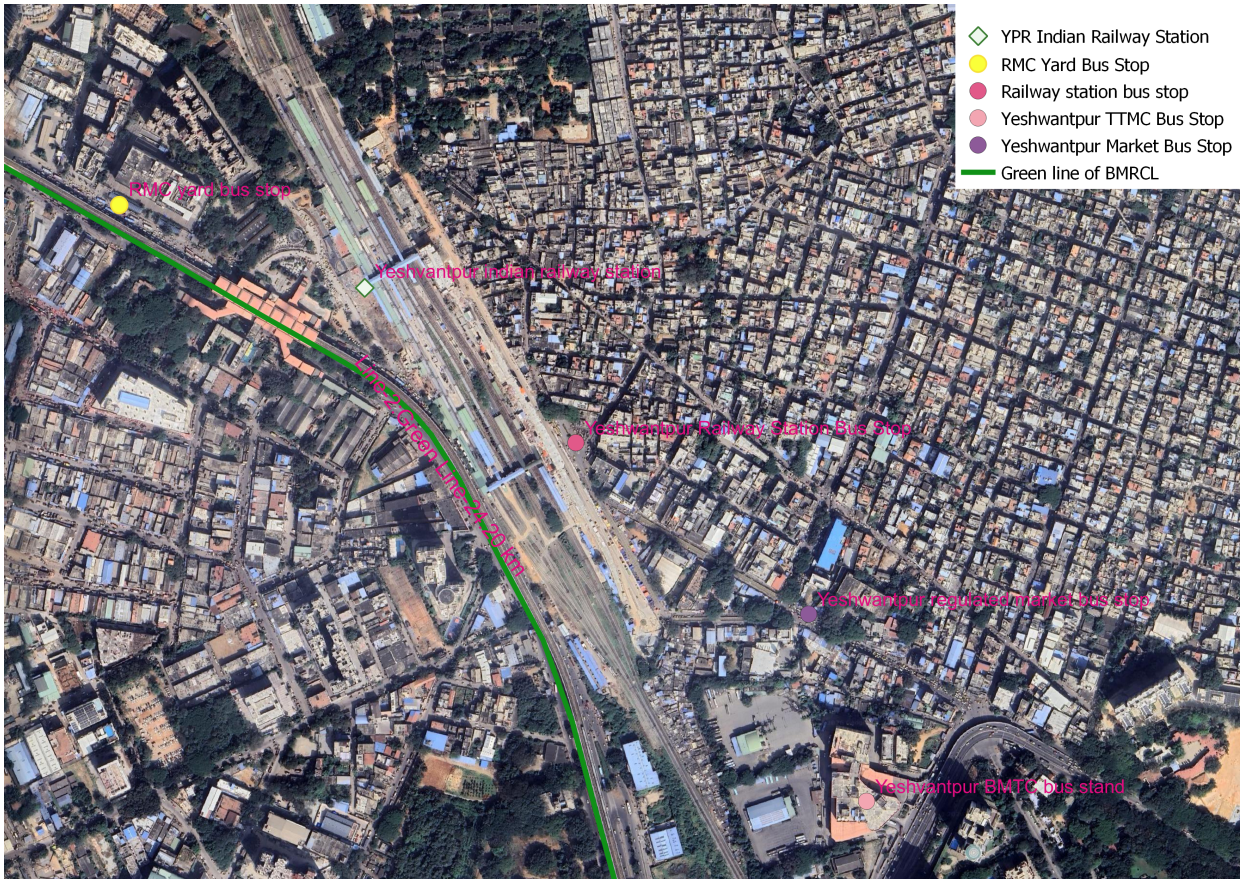


Figure 4.9: All the bus stops around YPR interchange area

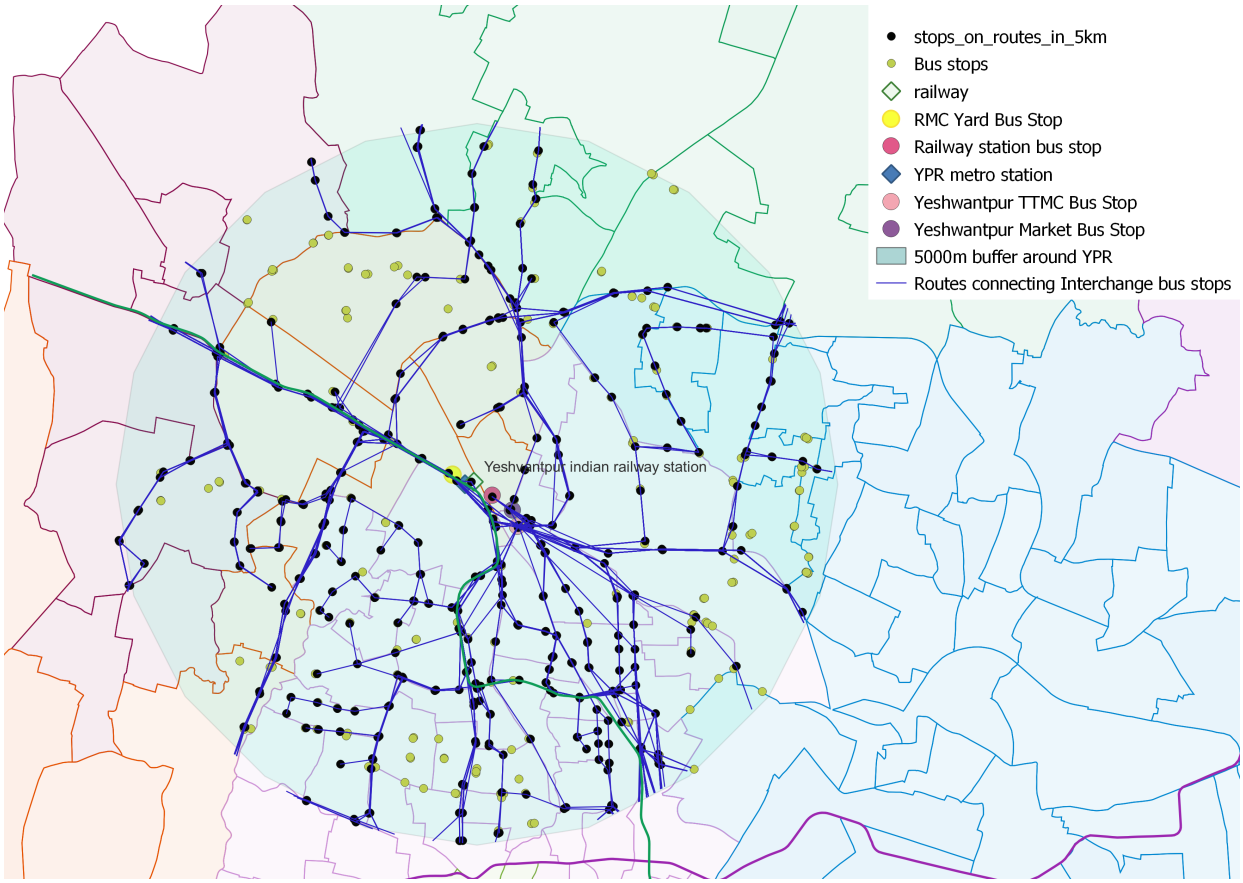


Figure 4.10: All the routes within the 5km buffer of interchange connecting all the interchange bus stops

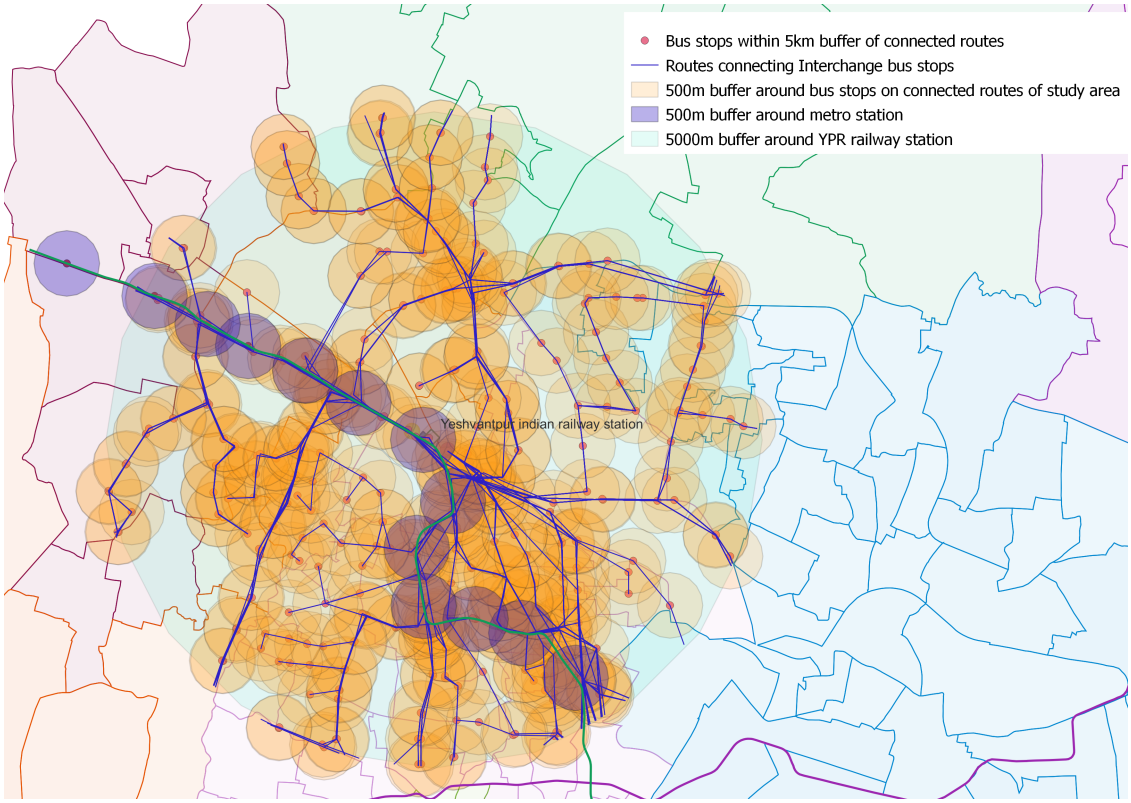


Figure 4.11: 500m buffer around bus stops and metro station

Chapter 5

Methodology

This chapter outlines the research design and simulation framework used to analyse first- and last-mile mode choice behaviour at Yeshwantpur railway station, Bengaluru. The study follows a three-stage structure — a base case simulation, a policy intervention, and a sensitivity analysis — as summarised in Figure 5.2. Each stage builds on the one preceding it, and together they form the basis for the policy recommendations presented in Chapter 7.

Why agent based modeling ? There are three major approaches in simulation modeling: system dynamics, discrete event, and agent based. While system dynamics is generally used for strategic level processes involving large aggregations, discrete event modeling is used to break down large-scale continuous process into discrete steps in order to reduce computational burden. On the other hand, agent-based models look into individual agent behavior. On one hand where system dynamics and discrete events approaches do not explicitly model agent behavior, agent-based approach do so in order to achieve global system behavior (Borshchev & Filippov 2004).

Available simulation tools for transportation engineers: A wide range of simulation tools are available to transportation engineers for analyzing traffic operations, travel behavior, and network performance at different spatial and temporal scales. PTV VISSIM and Aimsun are widely used microscopic traffic simulators for detailed modeling of vehicle interactions, signalized intersections, and corridor-level operations. For macroscopic planning and demand analysis, tools such as PTV VISUM support strategic transport planning, OD matrix estimation, and policy evaluation. Open-source platforms like SUMO provide flexible environments for large-scale traffic simulation and algorithm testing, while MATSim enables agent-based modeling of individual travel behavior and demand–supply interactions. In addition, AnyLogic supports hybrid simulation by integrating agent-based, discrete-event, and system-dynamics approaches, making it suitable for modeling multimodal transport systems and emerging mobility concepts. The choice of simulation tool depends on the study objectives, required level of detail, data availability, and computational complexity.

Why AnyLogic ? This study adopts AnyLogic as the primary simulation platform due to its strong support for agent-based modeling, which enables explicit rep-

resentation of individual travelers, vehicles, and their interactions within complex transportation systems and here also we are trying to build agent based simulation incorporating its built-in GIS mapping. So AnyLogic will be the ideal fit among all of these.

Figure 5.2 shows overall methodological framework of our study. The figure shows the steps to build the simulation and get the desired output.

5.1 Study Area and Simulation Boundary

The study focuses on the Yeshwantpur railway corridor in north-western Bengaluru. Yeshwantpur Junction is one of the city's major suburban and inter-city rail terminals, serving significant daily passenger volumes across commuter, intercity, and freight services. Its location at the intersection of the Namma Metro Green Line (Nagasandra–Silk Institute corridor) and a dense BMTC bus network makes it a strategically important multimodal interchange node.

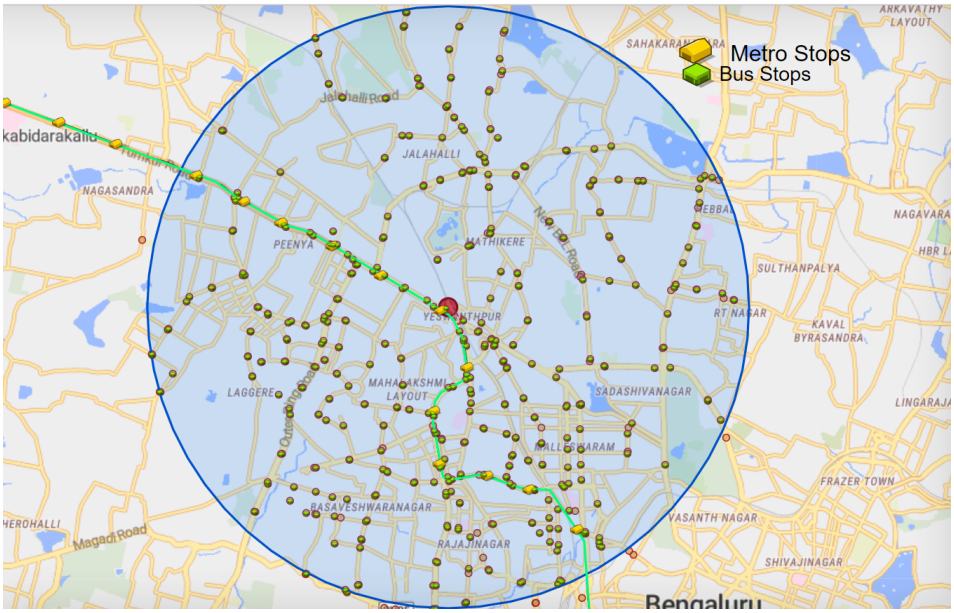


Figure 5.1: Defined simulation boundary and available transit network (on AnyLogic environment)

The simulation boundary is defined by a circular catchment zone centred on the station 5.1, with a radius calibrated to capture the realistic walk, auto-rickshaw, and feeder-bus shed of the station. Within this boundary, two spatial layers inform the model: a population-weighted origin grid representing residential density, and a transport network layer encoding bus stop locations, metro station positions, and road network geometry. All spatial data are projected to a common coordinate reference system to ensure consistency in distance and proximity calculations throughout the simulation.

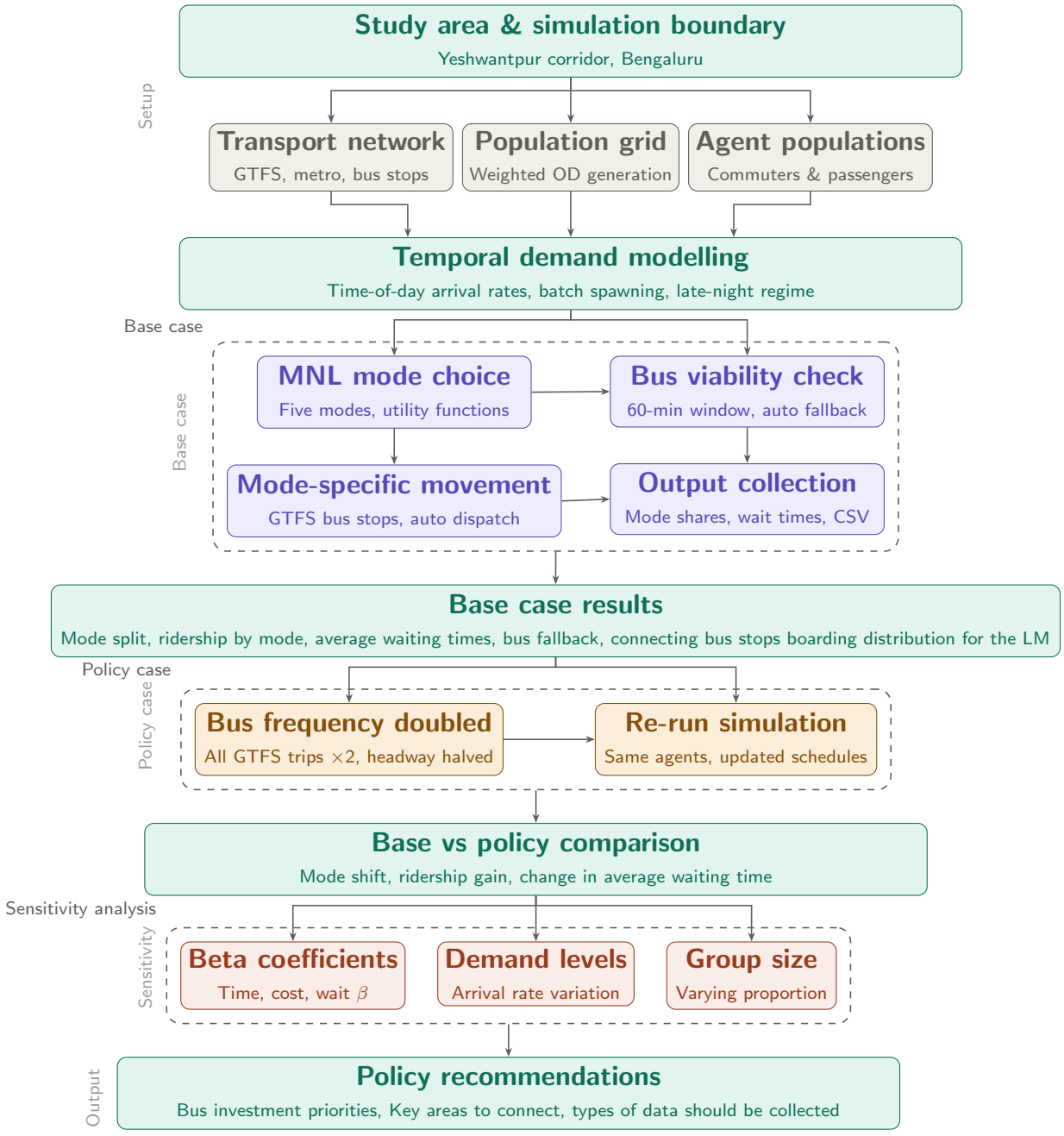


Figure 5.2: Overall research methodology flowchart.

5.2 Agent Population Design

The simulation models two conceptually distinct traveller populations and along with that several other agents (refer table 5.1). All of them instantiated as a separate agent class within the AnyLogic environment.

Table 5.1: Model Agents

Agent	Description
Main	Top-level agent where other agents reside
Passengers	Contains properties of last-mile passengers such as their last-mile destination location.
Commuters	Contains properties of first-mile passengers such as their origin location.
Auto	Contains properties of the IPT vehicles such as assigned first mile commuters, last mile passengers etc..
Bus	contains the properties of BMTC buses such as stop list, trip id.
BusStop	Contains the properties of BMTC bus stops such as lat, lon, stop id.

Commuters represent residents of the catchment zone who travel *toward* Yeshwantpur station to board a train. Their origins are drawn stochastically from a population-weighted grid covering the catchment area, and their destination is fixed at the station. Commuters are created throughout the simulation horizon according to a time-varying arrival rate that reflects observed peak and off-peak travel patterns.

Passengers represent travellers who have alighted at Yeshwantpur and are dispersing *away* from the station toward destinations elsewhere in the city. Their origins are fixed within the station catchment zone, and their destinations are generated using a population-weighted grid mechanism (in which grids having more population will have more chances to get selected as destination) that reflects the spatial distribution of trip attractors across Bengaluru.

Although the two populations share the same mode choice model structure, they differ in their origin-destination logic, mode availability checks, and the way bus waiting times are computed — differences that are detailed in subsequent sections. Both populations are assigned socioeconomic attributes (income group, gender, luggage weight, vehicle ownership, and group size) at the point of creation; the distributions governing these attributes are described in Section 5.5.

5.3 Origin–Destination Generation

5.3.1 Commuter Origins

Commuter origins are drawn from a discrete population grid covering the simulation catchment. Each grid cell is assigned a weight proportional to its estimated residential population density, derived from publicly available population data for Bengaluru. At agent creation, a cell is selected via inverse-CDF weighted sampling, and the

agent's origin coordinates are placed at the centroid of the selected cell. This ensures that commuter origins are spatially distributed in a manner consistent with where people actually live relative to the station, rather than being uniformly or randomly scattered.

5.3.2 Passenger Destinations

Passenger destinations are generated using the same weighted grid population mechanism: the grid here represents trip attraction rather than trip production, and the sampled coordinate becomes the agent's destination. The destination is generated at agent creation and remains fixed throughout the agent's lifecycle. The Euclidean distance between the station catchment origin and the sampled destination is then used to compute mode-specific travel times and costs for the MNL model.

5.3.3 Distance Computation

For both populations, the trip distance d (in kilometres) is computed as the straight-line Euclidean distance between origin and destination coordinates. While this is a simplification relative to network distance, it is applied consistently across all modes and therefore does not introduce systematic bias in the relative utility comparisons that drive mode choice.

5.4 Temporal Demand Modelling

Agent arrival rates vary continuously across the simulation horizon to reflect the diurnal rhythm of urban travel demand. The simulation is structured around a 24-hour clock, with the rate of agent creation governed by a time-of-day demand profile.

During normal operating hours, agents are spawned in batches at regular intervals, with the inter-arrival rate scaled to the time period. Morning and evening peak periods are assigned higher arrival rates, while midday and late-night periods are assigned lower rates. The demand profile is parameterised through a *lead hour* variable that allows the peak window to be shifted forward or backward in time, enabling the model to be calibrated to station-specific boarding and alighting patterns.

A distinct late-night regime applies between 00:00 and 05:59 for bus and between 00:00 and 4:59 for Namma Metro. During which both BMTC bus services and Namma Metro are operationally suspended. During this window, agent arrival rates are reduced substantially, and the vehicle ownership probabilities used in the MNL model are adjusted upward to reflect the composition of travellers active at these hours. Auto-rickshaws remain available throughout.

Group travel is handled through the group size attribute assigned at agent creation . All agents within a group share the same mode choice outcome, and the group size enters the MNL utility function as a penalty term for modes with practical capacity or comfort constraints.

5.5 Mode Choice Modelling: Multinomial Logit Framework

5.5.1 Overview

Mode choice in the simulation is governed by a Multinomial Logit (MNL) model, applied independently to two distinct traveller populations: *commuters*, who originate from residential zones within the study catchment and travel toward Yeshwantpur railway station, and *passengers*, who originate from the station area and travel to population-weighted destinations across the city. Both populations evaluate five competing modes — walking, auto-rickshaw, bus, metro, and private vehicle — and a mode is selected probabilistically based on computed utilities. The structure of the utility function is shared across populations, but several inputs are computed differently to reflect the distinct travel contexts of each group.

5.5.2 Agent Attribute Initialisation

At the moment of creation, each agent is assigned a set of socioeconomic and trip attributes that serve as independent variables in the MNL model. These attributes are drawn stochastically to reflect the heterogeneity observed in urban traveller populations in Bengaluru.

Gender

Gender is drawn from a Bernoulli distribution with a 60 per cent probability of being male ($g = 0$) and 40 per cent female ($g = 1$):

$$g \sim \text{Bernoulli}(0.40) \tag{5.1}$$

Income Group

Each agent is assigned to one of three income categories — low ($I = 0$), medium ($I = 1$), or high ($I = 2$) — based on a discrete cumulative distribution calibrated to Bengaluru's approximate household income distribution:

$$I = \begin{cases} 0 & \text{if } u < 0.45 \\ 1 & \text{if } 0.45 \leq u < 0.85 \\ 2 & \text{if } u \geq 0.85 \end{cases}, \quad u \sim U(0,1) \tag{5.2}$$

This yields approximately 45% low-income, 40% middle-income, and 15% high-income agents.

Luggage Weight and Category

Luggage weight is drawn from a triangular distribution with a minimum of 0 kg, mode of 3 kg, and maximum of 20 kg:

$$w_{\text{lug}} \sim \text{Triangular}(0,\ 3,\ 20) \tag{5.3}$$

The continuous weight is then discretised into a three-level luggage category used in the utility function:

$$L = \begin{cases} 1 & \text{if } w_{\text{lug}} \leq 3 \text{ kg} \\ 2 & \text{if } 3 < w_{\text{lug}} \leq 7 \text{ kg} \\ 3 & \text{if } w_{\text{lug}} > 7 \text{ kg} \end{cases} \tag{5.4}$$

Private Vehicle Ownership

Private vehicle ownership ($V \in \{0, 1\}$) is assigned via a Bernoulli-equivalent draw — $\texttt{uniform}(0, 1) < p$ — where the probability threshold p is conditioned on the agent's income group. Additionally, the threshold values are adjusted by time of day, with boosted ownership rates applied during late-night hours (12 AM–5 AM) to reflect the reduced availability of public transit alternatives.

During normal operating hours, low-income agents never own a private vehicle, while medium- and high-income agents are assigned ownership with probabilities of 35% and 75% respectively:

$$P(V = 1 \mid I) = \begin{cases} 0.00 & I = 0 \\ 0.35 & I = 1 \\ 0.75 & I = 2 \end{cases} \quad \text{(normal hours)} \tag{5.5}$$

During the late-night window (00:00–05:00), when bus and metro services are unavailable, these probabilities are adjusted upward to 10%, 45%, and 85% respectively, reflecting the reality that agents active during this period are more likely to be those who own or can access private transport.

Group Size

Both commuter and passenger agents travel in groups of one to six individuals. Group size is drawn from a triangular distribution:

$$n_{\text{grp}} = \begin{cases} 1 & \text{if } u < 0.30 \\ 2 & \text{if } 0.30 \leq u < 0.55 \\ 3 & \text{if } 0.55 \leq u < 0.75 \\ 4 & \text{if } 0.75 \leq u < 0.88 \\ 5 & \text{if } 0.88 \leq u < 0.95 \\ 6 & \text{if } u \geq 0.95 \end{cases}, \quad u \sim Triangular(0, 0.65, 1) \tag{5.6}$$

All agents within the same group share the same mode choice outcome.

5.5.3 Travel Time Computation

Travel time for each mode is computed from the origin-to-destination distance and an assumed average speed for that mode. For **commuters**, the relevant distance is from the agent's residential origin to Yeshwantpur railway station (d_C, in kilometres). For **passengers**, the distance is from the station catchment area to the agent's population-weighted destination (d_P, in kilometres). Assumed speeds are: walking

at 4 km/h, bus at 20 km/h, metro at 40 km/h, auto-rickshaw at 30 km/h, and private vehicle at 35 km/h.

Travel time for mode m is therefore:

$$TT_m = \frac{d}{\text{speed}_m} \times 60 \quad \text{(minutes)} \tag{5.7}$$

where the factor of 60 converts from hours to minutes.

5.5.4 Travel Cost Computation

Travel cost is computed using fare structures representative of Bengaluru's public and private transport market (all values in INR):

- **Walking:** $TC_{\text{walk}} = 0$
- **Auto-rickshaw:** A base fare of Indian Rupees (INR)25 plus INR12 per kilometre:
$$TC_{\text{auto}} = 25 + 12 \cdot d \tag{5.8}$$
- **Bus (BMTC):** A base fare of INR10 plus INR2 per kilometre:
$$TC_{\text{bus}} = 10 + 2 \cdot d \tag{5.9}$$
- **Metro (Namma Metro):** A slab-based fare with INR10 for trips up to 2 km, and INR5 per kilometre beyond that:
$$TC_{\text{metro}} = \begin{cases} 10 & d \leq 2 \text{ km} \\ 10 + 5 \cdot (d - 2) & d > 2 \text{ km} \end{cases} \tag{5.10}$$
- **Private vehicle:** A per-kilometre operating cost of INR7.5 (fuel and wear), with no base fare:
$$TC_{\text{private}} = 7.5 \cdot d \tag{5.11}$$

5.5.5 Waiting Time Computation

Waiting time is included as a separate disutility term for modes where it is non-trivial: auto-rickshaw, bus, and metro. It is computed at the time of agent creation using the current simulation clock.

Metro Waiting Time

Metro headway in the simulation is represented by a triangular distribution (minimum 2 min, mode 6 min, maximum 10 min), reflecting Namma Metro's off-peak and peak variation. Metro waiting time is set to `NA` during the late-night shutdown window (00:00–04:59), in which case it is treated as zero for utility calculation purposes.

$$wt_{\text{metro}} \sim \text{Triangular}(2,\ 6,\ 10) \text{ min} \tag{5.12}$$

Auto-Rickshaw Waiting Time

Auto waiting time combines a running mean of observed auto dispatch times from the simulation (updated dynamically as autos complete trips) with a stochastic component reflecting kerb-side hailing uncertainty:

$$wt_{\text{auto}} = \bar{w}_{\text{auto}}^{\text{running}} + \varepsilon, \quad \varepsilon \sim \text{Triangular}(10,\ 15,\ 20)\ \text{min} \tag{5.13}$$

where $\bar{w}_{\text{auto}}^{\text{running}}$ is the mean observed auto waiting time accumulated over all dispatched auto trips up to the current simulation time, converted to minutes.

Bus Waiting Time

Bus waiting time is computed from the live GTFS schedule data loaded into the simulation. For **commuters**, the nearest bus stop to the agent's origin is identified and the departure schedule of that stop is scanned within a 60-minute forward window. The simulation checks whether any upcoming departure serves at least one of seven Yeshwantpur interchange stops (representing the interchange nodes connecting to the station). The waiting time is the time in minutes until the next qualifying departure:

$$wt_{\text{bus}}^{C} = \frac{t_{\text{next_dep}} - t_{\text{now}}}{60}\ \text{min} \tag{5.14}$$

If no qualifying service is found within the 60-minute window, waiting time is set to 60 minutes. For **passengers**, the logic is symmetric but reversed: the nearest bus stop to the *destination* is identified, and interchange stops are queried for services that pass through or terminate near that destination. Both populations receive `NA` (treated as 0 in MNL) during the late-night bus shutdown (00:00–05:59).

5.5.6 Mode Availability Constraints

Before probabilities are computed, each mode's availability is checked against operational and spatial constraints:

- **Walking** is available only if the trip distance does not exceed 1,000 metres.
- **Bus** is available if the nearest bus stop is within 500 metres of the origin (for commuters) or destination (for passengers), and bus services are operating (i.e. outside the 00:00–05:59 shutdown window).
- **Metro** is available if the nearest metro station is within 500 metres of the relevant trip end, and metro services are operating (outside the 00:00–04:59 shutdown window).
- **Private vehicle** is available only to agents who are owing a private vehicle.
- **Auto-rickshaw** is always available.

Modes flagged as unavailable have their exponential utility term set to zero, effectively removing them from the choice set.

5.5.7 Utility Function Specification

The systematic utility V_m for mode $m \in \{\text{walk, auto, bus, metro, private}\}$ follows a linear-in-parameters specification:

$$V_m = \text{ASC}_m + \beta_{\text{time}} \cdot TT_m + \beta_{\text{cost}} \cdot TC_m + \beta_{\text{OVTT}} \cdot wt_m + \beta_{Im} \cdot I + \beta_{Lm} \cdot L + \beta_{\text{grp}} \cdot n_{\text{grp}} \quad (5.15)$$

Since, we did not conduct any survey to collect the data and estimate the parameter of the explanatory variables. We have taken some of the co-efficients from the related work done by Kou & Gu 2016 like $\beta_{luggage}$, β_{income} and Asc_m. Other co-efficients we have taken from our Choice Modeling course homework. All the co-efficients with their adopted values are tabulated in the table 5.2.

Table 5.2: MNL explanatory variable co-efficients

Parameter	Walking	Auto	Bus	Metro	Private
ASC(base/shift)	0.0	3.092	4.25	5.83	2.54-3.0
β_{time}	-0.036	-0.036	-0.036	-0.036	-0.036
β_{cost}	-0.726	-0.726	-0.726	-0.726	-0.726
β_{OVTT}	-0.058	-0.058	-0.058	-0.058	-0.058
β_{income}	NA	0.3059	-0.1858	-0.1851	0.0675
$\beta_{luggage}$	NA	-0.4562	-0.4738	-0.4323	-0.5748
β_{group}	NA	0.636	0.6025	2.857	0.857

5.5.8 Choice Probability and Mode Selection

Choice probabilities follow the standard logit formula. For a traveller n with available choice set C_n:

$$P(m \mid C_n) = \frac{\exp(V_m)}{\sum_{j \in C_n} \exp(V_j)} \quad (5.16)$$

A uniform random draw $r \sim U(0, 1)$ is then used to select the mode via cumulative probability comparison. A final feasibility check is applied after the draw: if the selected mode is subsequently found to be unavailable (e.g. private vehicle assigned to an agent without a vehicle), the fallback is reassignment to auto-rickshaw, which is always available. If the MNL draw assigns bus to an agent for whom no valid scheduled service can be confirmed within a 60-minute window of departure, that agent is also reallocated to auto, and the event is recorded as a bus-to-auto fallback in the simulation output.

5.5.9 Key Differences Between Commuter and Passenger MNL Inputs

Although the utility function structure is identical for both populations, the two groups differ in how several key inputs are computed, as summarised in Table 5.3.

Table 5.3: Key Differences in MNL Variable Computation Between Commuters and Passengers

Variable	Commuters	Passengers
Trip distance	From residential origin to Yeshwantpur railway station	From station catchment zone to a population-weighted random destination
Mode availability (proximity check)	Based on agent's current GIS location at origin	Based on destination coordinates
Bus waiting time	Scanned from nearest origin-side bus stop schedule; checks if departures serve any of seven Yeshwantpur interchange stops	Scanned from nearest destination-side bus stop; checks if any interchange trip serves the destination stop
Destination generation	Fixed (Yeshwantpur station)	Drawn via population grid weighted sampling mechanism.

5.6 Transport Network and GTFS Data Integration

The simulation integrates two primary transport network data sources: the General Transit Feed Specification (GTFS) dataset for BMTC bus services, and a point dataset of Namma Metro station locations along the Green Line.

5.6.1 GTFS Bus Schedule Integration

The BMTC GTFS feed provides stop-level departure schedules for all bus routes operating within the study area. At simulation initialisation, the full schedule is loaded into memory and indexed by stop identifier. During the simulation, each agent that has selected bus as its mode queries this live schedule to identify the next qualifying departure from the nearest bus stop within a 60-minute forward window. For commuters, a qualifying service is one that departs from a stop near the agent's origin and serves at least one of the designated Yeshwantpur interchange stops. For passengers, the query is reversed: the schedule is scanned for services that pass through or terminate near the agent's destination. This live schedule querying means that the simulation accurately reflects service availability at any given time of day, including the late-night shutdown window.

5.6.2 Metro Station Proximity

Metro availability is determined by a spatial proximity check: an agent has access to the metro if the nearest Green Line station is within 500 metres of the relevant trip end (origin for commuters, destination for passengers). Station coordinates are loaded from a GIS point dataset at initialisation. Metro headway is represented stochastically using a triangular distribution calibrated to Namma Metro's observed peak and off-peak service intervals.

5.6.3 Bus Stop Proximity

Similarly, bus availability requires the nearest bus stop to be within 500 metres of the relevant trip end. Bus stop locations are loaded from the GTFS stops file. If no stop falls within this threshold, or if no qualifying scheduled departure is found within the 30-minute window, the bus mode is marked unavailable for that agent.

5.7 Mode-Specific Movement and Operations

Once a mode has been selected, each agent is assigned a travel time and moves through the simulation accordingly through the defined statechart (refer figure 5.4).

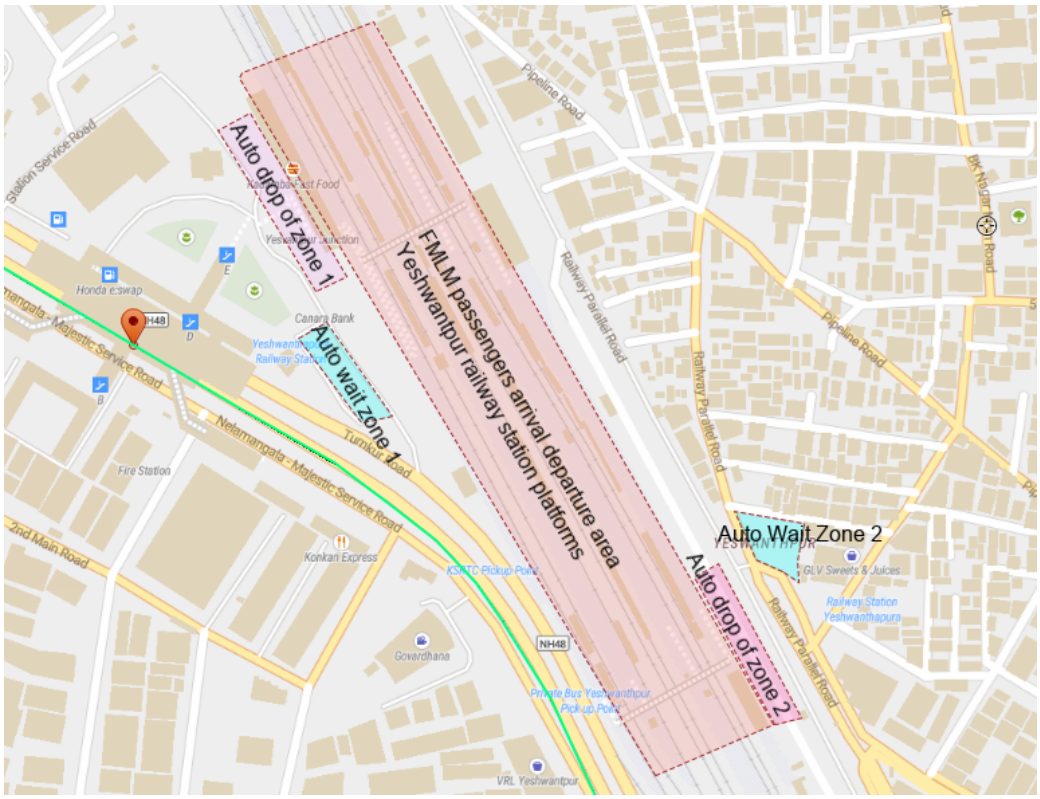


Figure 5.3: Defined station area, auto waiting and drop off zone at the interchange

The movement logic differs by mode. Initially, after generation, the agents will be at their origin (first mile) or at the station area (refer figure 5.3 for LM) and then based on their chosen mode they will move to their destination as follows-

Walking agents move directly from origin to destination at an assumed speed of 4 km/h. No additional waiting time is incurred.

Auto-rickshaw after assigned auto as the chosen mode through MNL model, agents will wait in the auto queue to get picked up by the auto. Then the auto will take the commuter to their destination (refer figure).

Bus agents wait at the nearest stop until the next qualifying GTFS-scheduled departure, then travel at an assumed average speed of 20 km/h. If no qualifying departure is found within 60 minutes, the agent falls back to other available modes (Section 5.8).

Metro agents move directly to the destination at a speed of 40 km/h.

Private vehicle agents travel at 35 km/h with no waiting time. This mode is only available to agents assigned vehicle ownership ($V = 1$) during attribute initialisation.

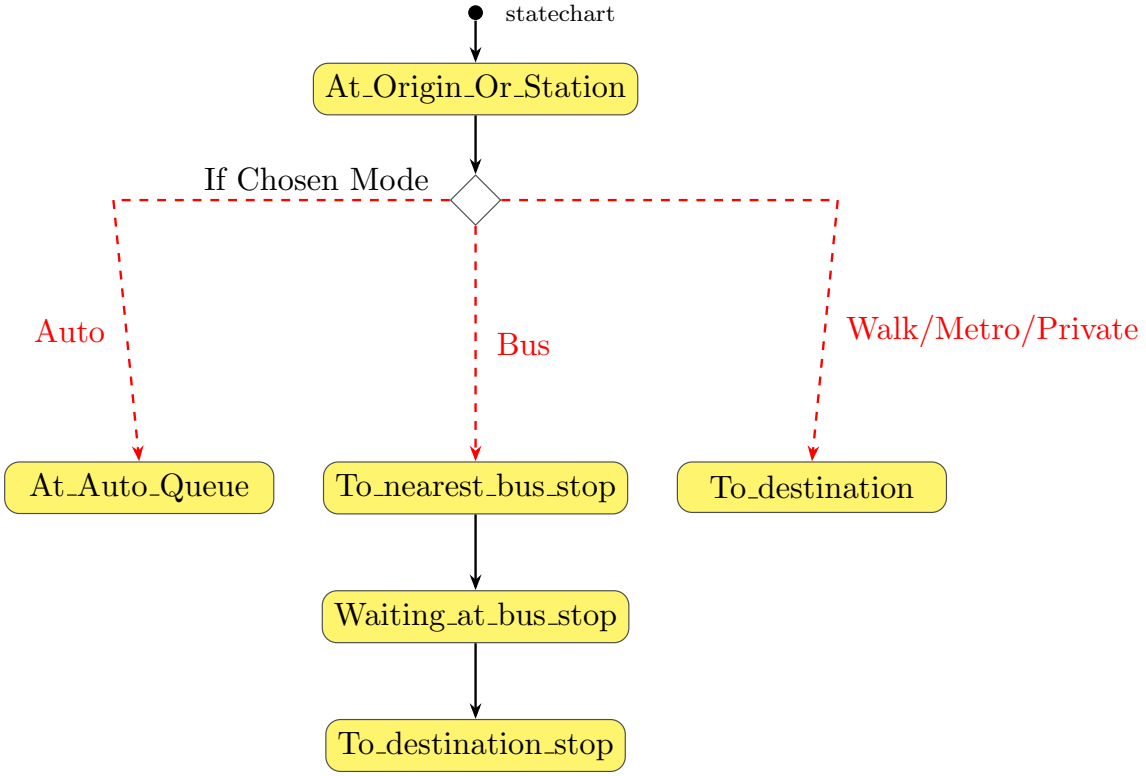


Figure 5.4: First and Last mile commuters state chart

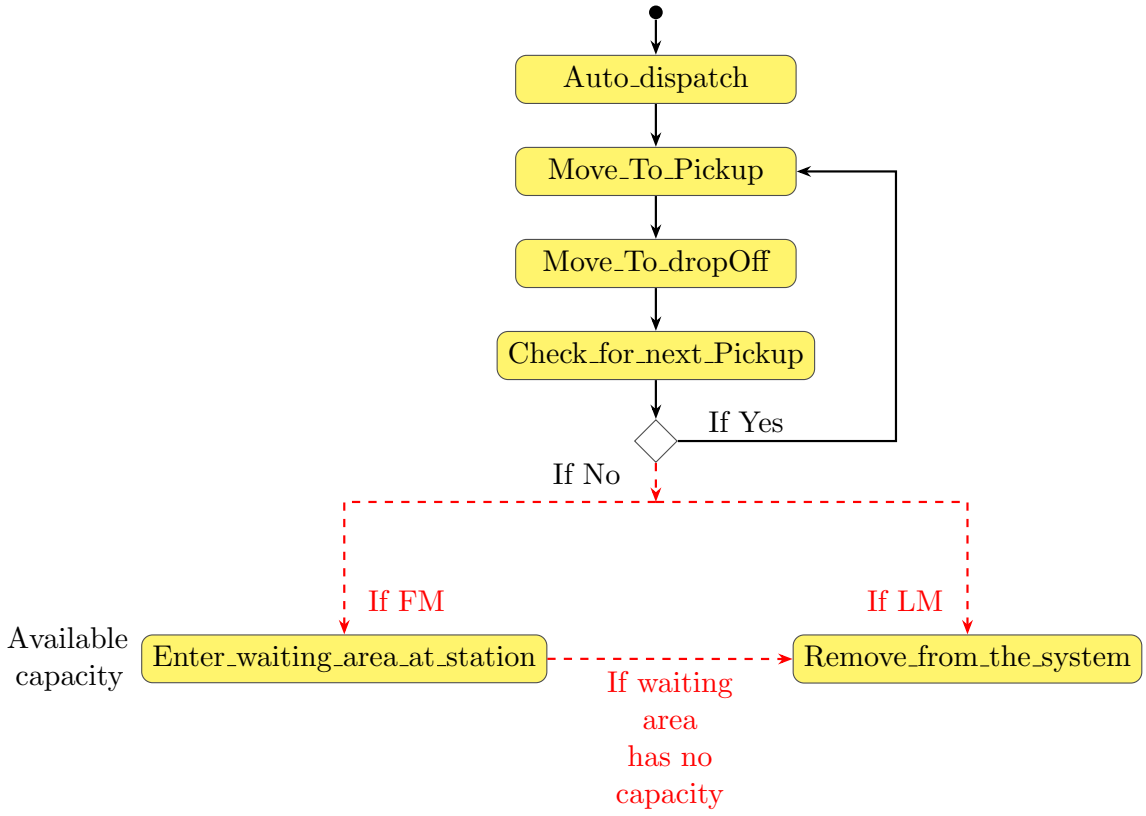


Figure 5.5: Auto state chart

Auto movement after the passenger enter into the auto queue For this study movement of the auto-rickshaw is defined with the following state chart (refer figure 5.5). Initially, autos will not be there in the system. As soon as any of FM or LM commuters enters into the auto queue, autos will generate within the 2km radius of that agent and go for pickup and consequently drop off of that agent. Now, this generation will continue till we hit the fleet capacity restriction. After drop off, if it is last mile commuter, then the auto will look for any first mile commuter queue within the 2km radius of its current location. If it finds one, go for pickup else, the auto will be removed from the system and nos of fleets in the system will be reduced by one. Now, on the other hand if it was first mile passenger i.e. currently the auto is in station area, then it will look for any last mile commuter who are currently waiting in the auto waiting area or any first mile commuter within 2km radius waiting in the auto queue. If it finds one, it will pick that up and go for the drop off of that first / last mile passenger. If, there is no one in the queue (both first mile or last mile), and the waiting area has available spot (since we are restricting at any time at max 100 autos can wait in the waiting area), then it will enter the waiting area. Else, it has to be removed from the system.

5.8 Bus Schedule Viability and Fallback Logic

A two-stage viability check governs whether an agent who has selected bus in the MNL draw can actually board a service. In the first stage, the nearest bus stop is identified and its proximity verified against the 500-metre threshold. In the second stage, the GTFS schedule is scanned for a qualifying departure within a 60-minute window from the current simulation time.

If assigned mode is bus through MNL model but no scheduled departure within next 60 minutes — the agent is reallocated to auto-rickshaw for the last mile commuters. But, for the first mile commuters they can be reallocated to any of the remaining available modes based on their utility. This fallback is recorded as a distinct event in the simulation output, allowing post-processing to quantify the volume of demand that bus services are failing to capture due to temporal service gaps. This metric is one of the key diagnostic outputs of the base case analysis.

5.9 Simulation Engine and Execution Parameters

The simulation is implemented in AnyLogic 8, a Java-based agent-based and discrete-event simulation platform. The simulation runs on a continuous time axis with a time unit of seconds, over a 24-hour horizon representing a full operating day. Agent creation, movement, and mode choice events are all processed in continuous time, meaning the model captures temporal interactions between demand and supply (such as auto waiting times growing during peak hours) without discretisation artefacts.

Each simulation run is executed as a single-run experiment with a fixed random seed for reproducibility. Output statistics are collected at the end of the run and exported to CSV and Excel formats for subsequent analysis. The model is implemented on a standard desktop machine; individual runs complete within a few minutes of wall-clock time.

5.10 Output and Data Collection

The simulation records the following output variables for each run:

- **Mode shares**: the number and proportion of agents selecting each of the five modes, reported separately for commuters and passengers.
- **Waiting times**: mean and distribution of waiting times by mode, capturing both the MNL-input waiting times and the realised waiting times after fallback reallocation.
- **Bus stop boarding counts**: the number of agents boarding at each GTFS stop, providing a spatially disaggregated picture of bus demand.
- **Fallback counts**: the number of agents reallocated from bus to auto-rickshaw due to schedule viability failures.
- **Auto dispatch statistics**: the running mean auto waiting time over the simulation horizon, reflecting the evolution of auto supply relative to demand.

These outputs are collected identically across the base case, policy case, and sensitivity analysis runs, enabling direct comparison.

5.11 Base Case

The base case represents current operating conditions at Yeshwantpur: observed BMTC bus schedules from the GTFS feed, existing metro service headways, and the income-stratified vehicle ownership rates described in Section 5.5. The base case simulation is run once with the full 24-hour demand profile and produces the mode split, ridership, waiting time, and fallback statistics that characterise the status quo. These results form the benchmark against which the policy intervention is evaluated.

5.12 Policy Case: Doubled Bus Frequency

The policy intervention tests the effect of a uniform doubling of bus service frequency across all BMTC routes operating within the study area. This is implemented by halving all inter-departure intervals in the GTFS schedule, effectively inserting an additional trip between every pair of consecutive scheduled trips on each route. All other simulation parameters — agent populations, socioeconomic attributes, MNL coefficients, metro headways, and auto supply — remain identical to the base case.

The primary hypothesis is that increased bus frequency will reduce bus waiting times, improve the competitive position of bus in the MNL utility comparison, and shift mode share away from auto-rickshaw toward bus. The policy case outputs are compared directly against the base case across all collected metrics to quantify the magnitude of this shift and identify which traveller groups benefit most.

5.13 Sensitivity Analysis

The sensitivity analysis examines the robustness of the base case and policy case results to uncertainty in three key model parameters.

Beta coefficients: The MNL coefficients for travel time ($\beta_{\text{luggageBus}}$), travel cost ($\beta_{\text{groupBus}}$), and waiting time ($\beta_{\text{wait}}$) are each perturbed by ± M% (Where M can take any value) around their assumed values, holding all other parameters constant. This tests whether the directional findings of the mode choice model are sensitive to the precise parameter estimates, or whether they hold robustly across a plausible range of valuations.

Demand levels: The agent arrival rate is scaled to 75%, 100%, and 150% and 200% of the base case profile to simulate lower- and higher-than-expected travel demand. This captures uncertainty in demand forecasts and tests whether the policy intervention remains effective under different demand scenarios.

Group size: It is always not necessary FMLM commuters will follow the same group size composition. To take that into account , we have varied it from base case and check its impact on both of weekdays and weekends corresponds to different fleet sizes.

Each sensitivity run varies one parameter at a time while holding all others at their base case values, following a one-at-a-time (OAT) design. Results are summarised as the change in mode share and mean waiting time relative to the base case, providing an elasticity-style measure of model sensitivity.

Chapter 6

Results and Discussion

6.1 Base Case

6.1.1 Key Performance Indicators Across Fleet Sizes

The base case simulation was run across four fleet sizes (F = 300, 400, 500, and 600) for both weekdays and weekends over a 24-hour horizon. Weekday commuter demand ranged from 14,000 to 14,800 agents, while last-mile passenger demand remained stable between 12,400 and 12,800. Weekend demand was somewhat lower and more uniform across fleet sizes. Auto-rickshaw emerged as the dominant mode for last-mile passengers at roughly 42–43% of trips, followed by bus at 36–40%. For first-mile commuters, the split was more balanced, with auto-rickshaw and bus together accounting for around 72–74% of trips. The group size distribution shows trips of size 3 dominating the population share at approximately 40%, followed by groups of 2 at 25% and groups of 4 at 21% — consistent across both first and last mile travellers. Auto-rickshaw waiting times showed notable variability, ranging from 9.2 minutes at F = 600 to 24.4 minutes at F = 400 for weekday commuters, suggesting a non-monotonic relationship with fleet size driven by demand fluctuations rather than supply alone. Last-mile passengers consistently experienced lower wait times than first-mile commuters, owing to the dedicated pool of idle autos stationed at the interchange. Bus fallback rates remained stable at 38–42% across all fleet sizes and day types. Average bus waiting time for last-mile passengers was approximately double that of first-mile commuters (19–21 min vs. 10–11 min), as last-mile passengers must skip buses arriving within 10 minutes of their station arrival, structurally inflating their average wait.

Table 6.1: Simulation KPI Summary for base case — All Fleet Sizes (24-hr horizon) For Weekdays

Metric	**F = 300**	**F = 400**	**F = 500**	**F = 600**
Demand (agents generated)				
Commuters (First Mile)	14,828	14,614	14,316	13,998
Passengers (Last Mile)	12,425	12,698	12,803	12,513
First-Mile Commuter Mode Split (%)				
Walking	2.5	2.5	2.8	2.7
Bus	33.2	32.9	32.7	34.0
Metro	9.5	9.9	9.6	8.9
Auto-Rickshaw	38.9	38.9	38.8	38.2
Private	15.9	15.8	16.1	16.2
Last-Mile Passenger Mode Split (%)				
Walking	3.3	3.5	3.5	3.4
Bus	39.6	39.7	39.1	36.0
Metro	4.2	4.3	4.2	4.9
Auto-Rickshaw	42.7	42.2	43.1	43.4
Private	10.1	10.3	10.0	12.3
Auto Waiting Time Commuters (min)				
Average	19.88	24.36	20.00	9.22
Min / Max	0.00 / 40.64	0.00 / 33.88	0.00 / 28.77	0.00 / 36.48
Auto Waiting Time Passengers (min)				
Average	14.38	17.61	11.86	6.41
Min / Max	0.04 / 47.63	0.02 / 33.68	0.04 / 27.68	0.02 / 29.47
Bus Schedule Fallback				
Commuters (% of MNL-Bus)	41.2	41.8	41.9	40.2
Passengers (% of MNL-Bus)	38.3	38.1	39.4	37.9
Avg bus wait commuters (min)	10.80	11.10	11.06	10.80
Avg bus wait passengers (min)	20.64	20.03	20.38	19.76

Table 6.2: Simulation KPI Summary for base case — All Fleet Sizes (24-hr horizon) For Weekends

Metric	**F = 300**	**F = 400**	**F = 500**	**F = 600**
Demand (agents generated)				
Commuters (First Mile)	13,293	13,493	13,616	13,501
Passengers (Last Mile)	13,353	13,352	13,011	13,126
First-Mile Commuter Mode Split (%)				
Walking	3.2	3.0	2.8	2.5
Bus	38.3	37.6	37.3	36.9
Metro	10.1	9.8	10.3	11.3
Auto-Rickshaw	35.2	36.0	35.9	35.9
Private	13.2	13.6	13.7	13.4
Last-Mile Passenger Mode Split (%)				
Walking	3.4	3.5	3.4	3.5
Bus	39.9	39.6	39.9	39.8
Metro	4.2	4.2	4.3	4.2
Auto-Rickshaw	42.2	42.4	41.9	42.0
Private	10.3	10.3	10.5	10.5
Auto Waiting Time Commuters (min)				
Average	25.52	19.26	21.40	2.47
Min / Max	0.00 / 40.16	0.03 / 38.87	0.01 / 33.31	0.00 / 29.80
Auto Waiting Time Passengers (min)				
Average	15.08	16.76	14.29	8.10
Min / Max	0.01 / 39.87	0.01 / 33.66	0.01 / 28.95	0.01 / 25.33
Bus Schedule Fallback				
Commuters (% of MNL-Bus)	38.9	39.4	39.2	39.5
Passengers (% of MNL-Bus)	37.3	37.6	36.7	37.0
Avg bus wait commuters (min)	10.45	11.15	10.83	10.74
Avg bus wait passengers (min)	18.97	18.84	18.99	18.80

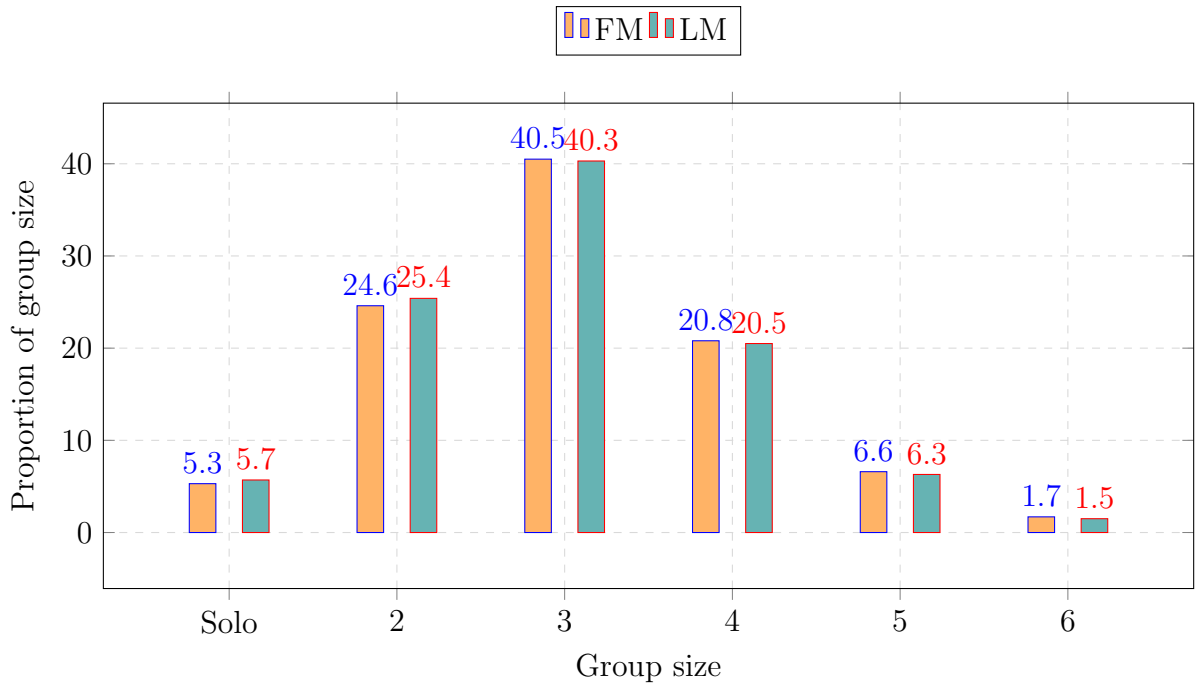


Figure 6.1: Group size variation for the first mile and last mile passengers

6.2 Policy Case

6.2.1 Key Performance Indicators Across Fleet Sizes

The policy case introduces a single operational intervention — doubling the bus frequency — while keeping all other model parameters identical to the base case, including demand generation, mode choice coefficients, and group size proportions. Demand levels remained comparable to the base case, with weekday first-mile commuters ranging from 14,026 to 14,580 agents and last-mile passengers between 12,207 and 12,982. Mode split patterns were broadly similar, with auto-rickshaw continuing to dominate last-mile trips at 44–45% — marginally higher than the base case — and bus share settling at 35–38%. This slight shift away from bus is explained by the increased fallback rate: with higher frequency, more passengers are MNL-assigned to bus, but a greater proportion encounter buses arriving within the 10-minute exclusion window at the key stops, forcing fallback to auto. The most notable improvement is in average bus waiting time. First-mile commuters saw waiting times drop to 8.4–9.0 minutes on weekdays from 10.8–11.1 minutes in the base case — a reduction of roughly 2 minutes directly attributable to doubled frequency. Last-mile passenger bus wait times similarly reduced from 19–21 minutes to 17–18 minutes. Auto waiting times, however, showed no consistent improvement and in several scenarios increased relative to the base case, driven by marginally higher auto demand from bus fallback. Overall, doubling bus frequency delivers a clear reduction in bus waiting times for both traveller groups, but its impact on mode share remains limited, pointing to stop-level scheduling constraints as the binding barrier for last-mile bus uptake.

Table 6.3: Simulation KPI Summary for policy case — All Fleet Sizes (24-hr horizon) For Weekdays

Metric	**F = 300**	**F = 400**	**F = 500**	**F = 600**
Demand (agents generated)				
Commuters (First Mile)	14,580	14,026	14,518	14,127
Passengers (Last Mile)	12,445	12,982	12,330	12,207
First-Mile Commuter Mode Split (%)				
Walking	2.6	2.7	2.8	2.7
Bus	35.7	34.5	35.4	35.0
Metro	9.5	9.5	10.1	9.5
Auto-Rickshaw	36.7	37.4	36.6	37.2
Private	15.5	15.9	15.1	15.6
Last-Mile Passenger Mode Split (%)				
Walking	3.4	3.3	3.4	3.5
Bus	37.9	38.2	37.8	35.0
Metro	4.3	4.2	4.2	4.7
Auto-Rickshaw	44.4	44.1	44.5	44.5
Private	10.0	10.2	10.0	12.3
Auto Waiting Time Commuters (min)				
Average	19.71	21.03	26.13	22.54
Min / Max	0.01 / 43.54	0.00 / 37.47	0.00 / 29.39	0.00 / 29.50
Auto Waiting Time Passengers (min)				
Average	10.04	12.18	14.36	14.22
Min / Max	0.01 / 42.96	0.02 / 35.79	0.03 / 33.10	0.04 / 29.47
Bus Schedule Fallback				
Commuters (% of MNL-Bus)	38.4	39.7	37.7	39.2
Passengers (% of MNL-Bus)	41.8	41.5	42.2	40.4
Avg bus wait commuters (min)	8.52	8.95	8.43	8.58
Avg bus wait passengers (min)	17.75	17.29	18.09	17.06

Table 6.4: Simulation KPI Summary for policy case — All Fleet Sizes (24-hr horizon) For Weekends

Metric	**F = 300**	**F = 400**	**F = 500**	**F = 600**
Demand (agents generated)				
Commuters (First Mile)	13,482	13,346	13,377	13,478
Passengers (Last Mile)	13,347	12,981	13,597	13,801
First-Mile Commuter Mode Split (%)				
Walking	3.0	2.7	2.9	2.4
Bus	39.1	39.8	39.8	40.0
Metro	10.1	10.4	10.3	10.1
Auto-Rickshaw	34.5	34.0	33.8	33.8
Private	13.2	13.2	13.1	13.7
Last-Mile Passenger Mode Split (%)				
Walking	3.5	3.4	3.4	3.4
Bus	38.9	37.8	38.6	38.7
Metro	4.3	4.2	4.2	4.2
Auto-Rickshaw	42.8	44.2	43.3	43.5
Private	10.5	10.4	10.5	10.2
Auto Waiting Time Commuters (min)				
Average	28.22	22.17	23.44	25.79
Min / Max	0.00 / 40.11	0.01 / 36.15	0.01 / 34.60	0.00 / 30.71
Auto Waiting Time Passengers (min)				
Average	15.30	17.95	18.21	16.49
Min / Max	0.01 / 47.82	0.01 / 38.46	0.01 / 24.24	0.01 / 23.17
Bus Schedule Fallback				
Commuters (% of MNL-Bus)	37.9	37.1	36.9	37.1
Passengers (% of MNL-Bus)	39.4	41.0	40.1	40.2
Avg bus wait commuters (min)	8.25	8.27	8.43	8.48
Avg bus wait passengers (min)	17.01	16.83	16.89	16.81

6.3 Comparative Analysis

6.3.1 Average Auto-Rickshaw waiting time

For base case and policy case there is not much improvement in terms of average waiting time for both of first mile and the last mile commuters. In most of the cases it got increased. The main reasons could be the nos of commuters using the service and also their arrival in the system. For example except fleet size 400 scenario case for all other case more first and last mile commuters have generated which means more auto users. That in turns leads to more waiting time.

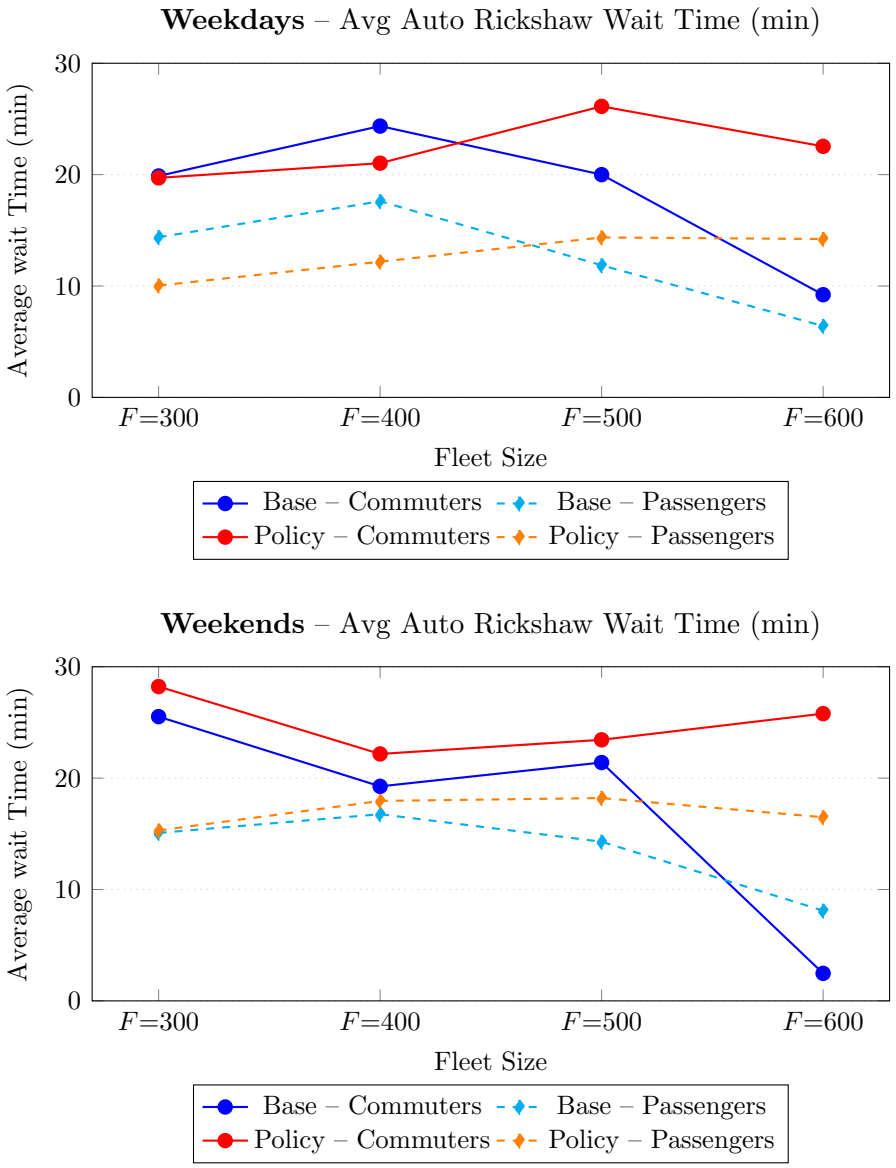


Figure 6.2: Comparison of average Auto-Rickshaw wait time between base case and policy case

Also, first mile commuters are facing more average waiting time as compared to last mile commuters for both of the base case and policy case. The main reasons could be last mile passengers are peaking before the first mile passengers and thus

most of the autos are occupied with last mile passengers and after that only they are going to pickup for the first mile passengers.Also, in this study we did not keep any provisions for first mile passengers like for the last mile passengers there are some 100 idle autos in the station area if there is no first (within 2km of the station area) or last mile passengers in the queue block. So, for first mile passengers - they cant get immediate auto since it will come from within 2km area. But for last mile passengers they got the privilege of having instant auto since there are always around 100 idle autos waiting at the station area.

6.3.2 Bus fallback analysis

For both of base case and policy case scenarios, for both of weekdays and weekends across all fleet sizing scenarios, we can see different trends for the first mile commuters and last mile commuters which is discussed below.

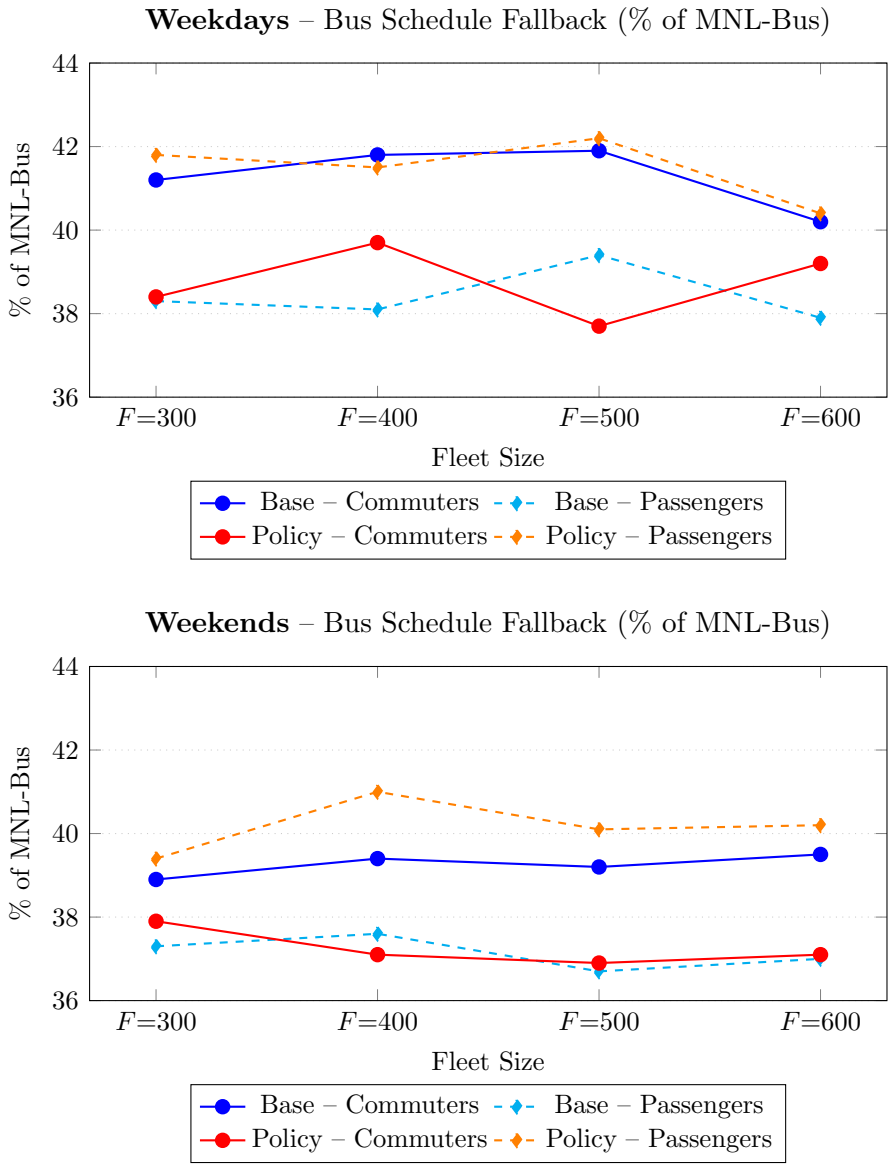


Figure 6.3: Bus schedule fallback comparison between base case and policy case

Trends for the first mile commuters : For first mile commuters we can see due to increased frequency in the policy case results in less waiting time and more commuters choosing bus both for weekdays and weekends. In the figure 6.4 we have shown the trend for the weekdays but same can be observed during weekends also . Since they can take the immediate bus so the fallback has reduced across all fleet sizes for both of weekdays and weekends as we can see from the 6.3.

For the last mile passengers : For the last mile passengers also, similarly MNL mode share increases. But, they can take it if it is within next 1hr and also for those 4 bus stops they have to skip it if is before next 10 mins. And we can see across all fleet sizes and for both of weekdays and weekends fallback increases as compared to base case scenario. The most possible reasons could be-

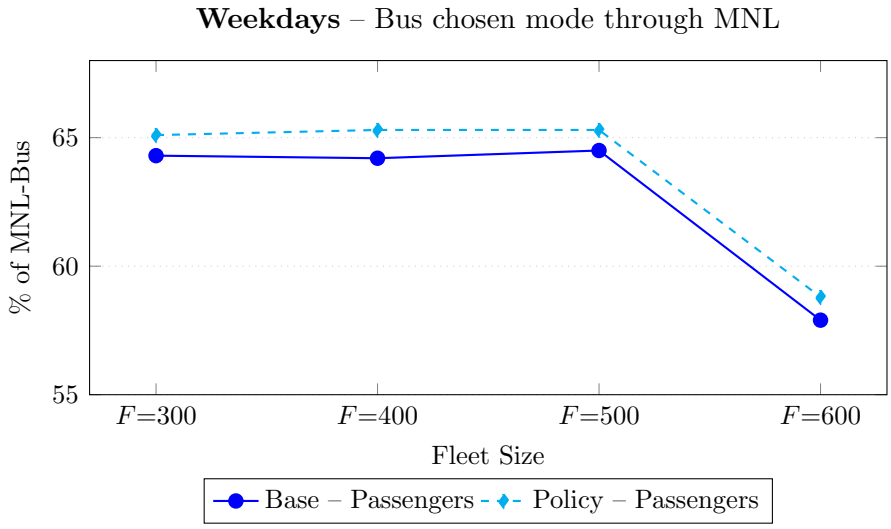


Figure 6.4: MNL mode share for base case and policy case for weekdays

- Bus might be after 10 minutes from any of those 4 stops (stop id 21288,24016,21131 and 21132). So according to MNL bus is the best mode, but after mode choice they have to fallback to auto if the next bus is not within 1hr beyond that 10 minutes.

6.3.3 Average Bus waiting time

For both policy case and base case we can see a clear difference of average waiting (around 10 mins refer figure 6.5) time between the first mile and last mile commuters for both of weekdays and weekends. The main reason behind this, for first mile commuters they will take the immediate next bus from their nearby stops to one of those 7 connecting bus stops at the interchange area. So, it is assumed that they are aware about the bus timings from their nearby stops and since it is within 500 m from their origin, so they can make it to the bus stop before it arrives.

But, for the last mile commuters- they are looking for buses to their destinations nearby bus stops from those one of the 7 connecting bus stops. But, if the bus is within next 10 mins they cant make it except for the bus stop id 27535, 23334

Table 6.5: Percentage share of the bus stops for the last mile passengers successful boarding

Stop Id	Fleet size scenario	
	Weekend & 500	Weekday & 600
21131	5.4%	4.9%
21132	13.8%	14.8%
27535	3.7%	3.8%
27453	3.6%	3.3%
23334	0.8%	1.1%
24016	21.1%	20.1%
21288	51.7%	52.0%

and 27453. And from the model obtained results it is found that most of the last mile passengers who are getting connecting buses - are mostly from the stop id 21131,21132, 21288 and 24016 as you can see from the table 6.5 for some of the base case fleet size scenarios. Since, for them have to skip the bus within 10 mins from their arrival at the station area, so they have to take the next bus which is after 10 mins and thus for them bus waiting time is beyond 10 mins and taking the average away from 10 mins.

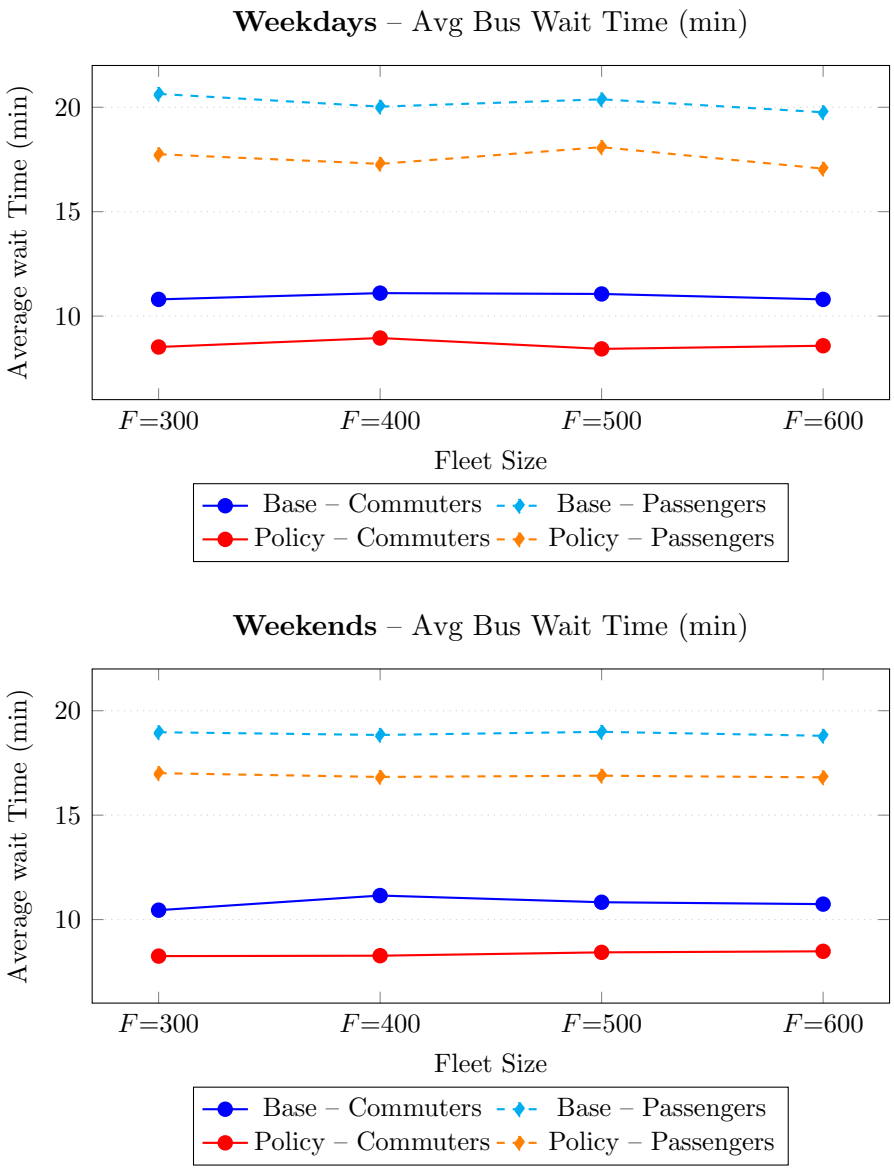


Figure 6.5: Average bus waiting time comparison between base case and policy case

6.3.4 Mode share

For both the base case and policy case and also for both weekdays and weekends across all fleet size scenarios, we can see last mile passengers are more prone towards using autos as compared to the first mile commuters (as you can see from the figure 6.6 for some of the cases). The most possible reasons are as follows-

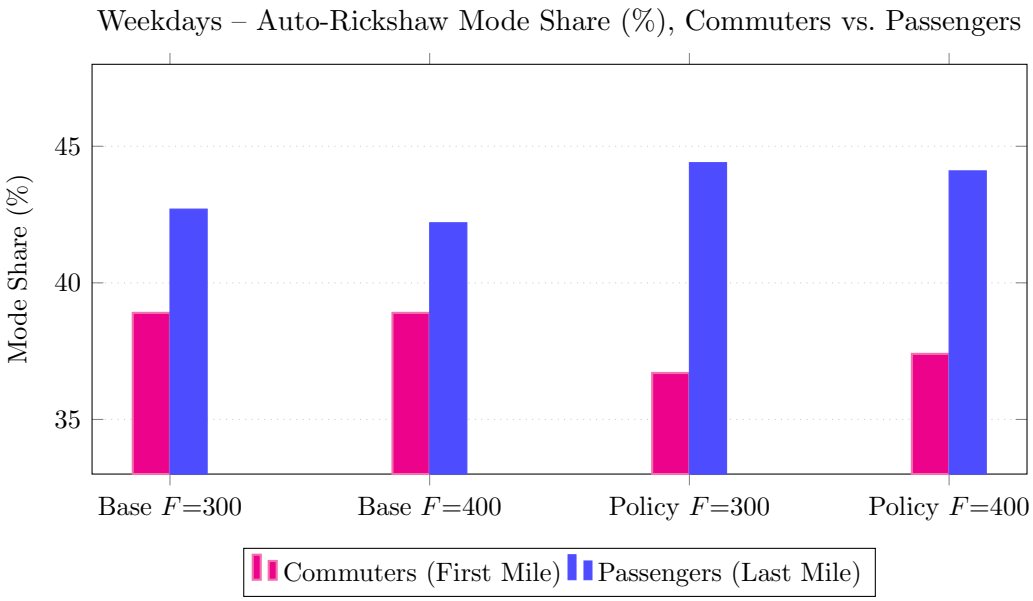


Figure 6.6: Auto-rickshaw mode share comparison between the first and last mile commuters

- Average waiting time for auto is more for the first mile commuters.
- Average bus waiting time is more for the last mile commuters.
- Last mile commuters are bound to take auto if they are falling back from bus.
- First mile commuters can take any of auto, metro or private vehicle if they are falling back from bus and from the tables we can see first mile commuters are using more metro and private vehicles as compared to the last mile commuters.

6.4 Sensitivity Analysis

6.4.1 Group size variation

Since, it is always not expected the group formation among the first and the last mile commuters will be same. To see how group formation heterogeneity impacts the key performance indicators of our model we have increased the proportion of solo and group of 2 travelers proportion from the base case which results in reduction in all other group size proportion for both the first and the last mile commuters (refer figure 6.1.1 and 6.4.1).

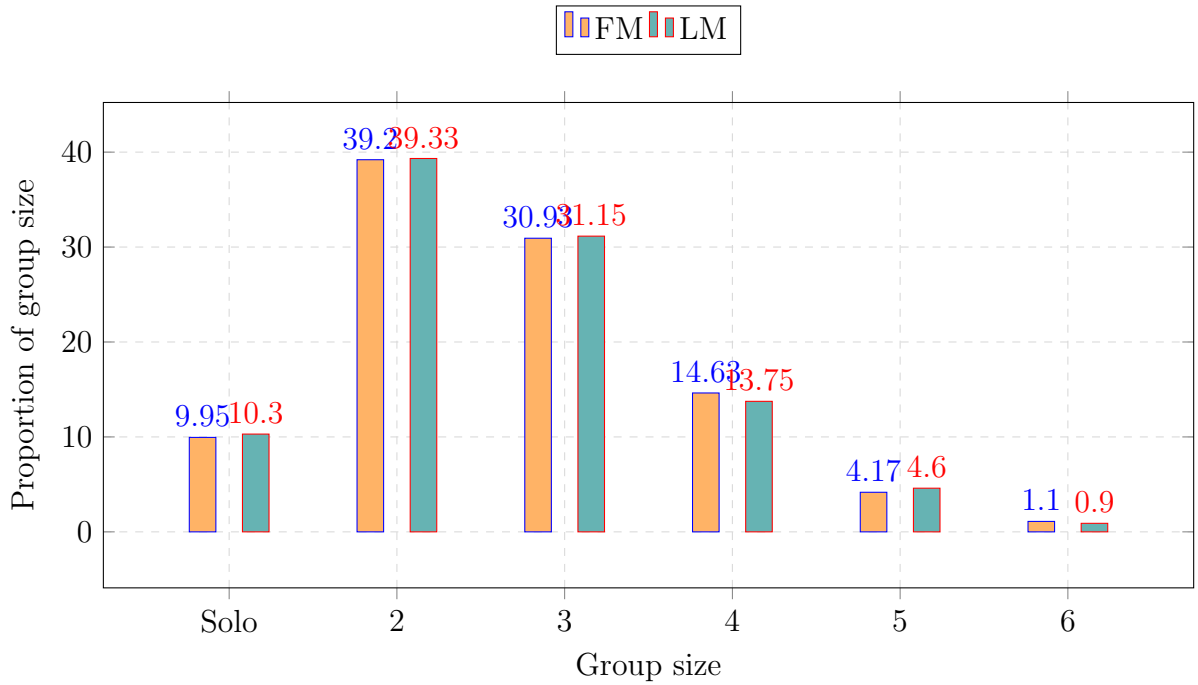


Figure 6.7: Group size variation for the first mile and last mile passengers

Now for both weekdays and weekends, we have run the model for 300 and 400 fleet size variation only . All the observed results are tabulated on table 6.6 and 6.7.

Major indicators performance : From the tables 6.6 and 6.7 we can see average bus waiting time increases for both of the first and last mile commuters due to increase bus share except for the first mile commuters on weekends.

For the average auto waiting time : Both on weekdays and weekends, since the mode share for the first mile commuters for auto is reducing thus average waiting time is also reducing.
On the contrary, the mode share for the last mile commuters for auto is increasing but the average waiting time is still decreasing.

Table 6.6: Simulation KPI Summary for group size variation — All Scenarios (24-hr horizon) for Weekdays

Metric	Base case		Policy case	
	300	**400**	**300**	**400**
Demand (agents generated)				
Commuters (First Mile)	14,828	14,614	14,408	14,553
Passengers (Last Mile)	12,425	12,698	12,643	12,400
First-Mile Commuter Mode Split (%)				
Walking	2.5	2.5	3.4	3.4
Bus	33.2	32.9	34.4	33.9
Metro	9.5	9.9	9.3	9.2
Auto-Rickshaw	38.9	38.9	37.6	37.6
Private	15.9	15.8	15.4	15.9
Last-Mile Passenger Mode Split (%)				
Walking	3.3	3.5	3.3	3.4
Bus	39.6	39.7	40.1	40.0
Metro	4.2	4.3	4.0	4.1
Auto-Rickshaw	42.7	42.2	43.0	42.7
Private	10.1	10.3	9.6	9.8
Auto Waiting Time Commuters (min)				
Average	19.88	24.36	19.87	22.10
Min / Max	0.00 / 40.64	0.00 / 33.88	0.00 / 45.20	0.00 / 40.37
Auto Waiting Time Passengers (min)				
Average	14.38	17.61	13.82	16.96
Min / Max	0.04 / 47.63	0.02 / 33.68	0.07 / 45.41	0.04 / 35.35
Bus Schedule Fallback				
Commuters (% of MNL-Bus)	41.2	41.8	38.7	39.9
Passengers (% of MNL-Bus)	38.3	38.1	39.1	39.0
Avg bus wait commuters (min)	10.80	11.10	11.53	11.27
Avg bus wait passengers (min)	20.64	20.03	20.79	20.40

Table 6.7: Simulation KPI Summary for group size variation — All Scenarios (24-hr horizon) for Weekends

Metric	Base case		Policy case	
	300	**400**	**300**	**400**
Demand (agents generated)				
Commuters (First Mile)	13,293	13,493	13,356	13,298
Passengers (Last Mile)	13,353	13,352	13,510	13,091
First-Mile Commuter Mode Split (%)				
Walking	3.2	3.0	3.1	2.7
Bus	38.3	37.6	37.3	37.4
Metro	10.1	9.8	10.3	10.0
Auto-Rickshaw	35.2	36.0	35.7	36.0
Private	13.2	13.6	13.6	13.9
Last-Mile Passenger Mode Split (%)				
Walking	3.4	3.5	3.4	3.3
Bus	39.9	39.6	40.2	40.6
Metro	4.2	4.2	4.3	4.2
Auto-Rickshaw	42.2	42.4	41.8	41.7
Private	10.3	10.3	10.3	10.1
Auto Waiting Time Commuters (min)				
Average	25.52	19.26	18.78	12.08
Min / Max	0.00 / 40.16	0.03 / 38.87	0.02 / 28.54	0.01 / 39.26
Auto Waiting Time Passengers (min)				
Average	15.08	16.76	9.41	16.73
Min / Max	0.01 / 39.87	0.01 / 33.66	0.03 / 31.13	0.03 / 31.14
Bus Schedule Fallback				
Commuters (% of MNL-Bus)	38.9	39.4	39.1	39.9
Passengers (% of MNL-Bus)	37.3	37.6	37.1	36.7
Avg bus wait commuters (min)	10.45	11.15	11.13	11.11
Avg bus wait passengers (min)	18.97	18.84	18.84	18.67

6.4.2 Group Co-efficient variation

Since, we have made huge assumption in terms of the co-efficients of MNL model independent variables parameters, so it is better to see how the variation of those co-efficients can impact our model performance for this case study. For the first we have taken luggage co-efficeint for the bus mode i.e. $\beta_{luggageBus}$. In the following table we have shown the varaiation in key performance indicators only for one fleet size scenario and for weekdays. Though we can obtain some variation for other fleet size scenarios and on weekends also. But due to time constrains, so far we are able to run it for one scenario and weekdays.

For group Bus co-efficeint variation

The bus group coefficient ($\beta_{groupbus}$) was varied across four values (0.3, 1.5, 3.5, and 6.0) (refer table 6.8) at fleet size 600 on weekdays. As the coefficient increases, bus mode share rises modestly for both first-mile commuters (35.5% to 40.9%) and last-mile passengers (38.0% to 40.1%). The gain is largely drawn from metro, whose share for first-mile commuters drops from 8.4% to 4.4%, while auto-rickshaw share remains virtually unchanged at 38–39% for commuters and 44–46% for passengers — confirming that auto demand is inelastic to bus attractiveness in this setting. Bus fallback rates and average waiting times show negligible variation across the range, indicating that scheduling constraints govern fallback behaviour independently of MNL utility. Overall, while the model responds directionally as expected, the moderate magnitude of mode shift suggests that improving bus attractiveness through group preference alone yields limited ridership gains without complementary improvements in schedule reliability.

For luggage group co-efficient variation

The luggage bus coefficient ($\beta_{luggageBus}$) was varied across four values (1.0, 0.4738, +0.4738, and +1.0) (refer table 6.9) at fleet size 600 on weekdays, where negative values reflect a penalty for travelling with luggage on bus and positive values reflect a reduced deterrence. As the coefficient moves from negative to positive, bus mode share increases for both first-mile commuters (33.9% to 39.6%) and last-mile passengers (36.8% to 39.6%), with the corresponding decline concentrated in metro share for commuters (8.7% to 5.2%) and metro and private for passengers. Auto-rickshaw share again remains largely stable across the range at 38–39% for commuters and 44–46% for passengers, reinforcing the finding that auto demand is insensitive to changes in bus utility parameters. Auto waiting times for commuters show a notable increasing trend from 8.2 to 30.6 minutes as the coefficient rises, likely reflecting higher auto demand from induced bus fallback. Bus fallback rates and average bus waiting times remain broadly stable, consistent with the pattern observed in the group coefficient sensitivity. Overall, the luggage coefficient has a meaningful influence on bus attractiveness, particularly for last-mile passengers who are more likely to be carrying luggage, but schedule reliability remains the dominant constraint on actual bus boarding.

Table 6.8: Simulation KPI Summary — Bus Group co-efficients sensitivity (Fleet = 600, Weekday)

Metric	**0.3**	**1.5**	**3.5**	**6.0**
Demand (agents generated)				
Commuters (First Mile)	14,336	14,141	14,543	14,543
Passengers (Last Mile)	13,001	12,377	12,835	12,594
First-Mile Commuter Mode Split (%)				
Walking	3.5	3.1	3.1	2.2
Bus	35.5	35.9	38.3	40.9
Metro	8.4	8.1	5.7	4.4
Auto-Rickshaw	37.5	38.3	38.6	38.4
Private	15.0	14.6	14.2	14.1
Last-Mile Passenger Mode Split (%)				
Walking	3.2	3.5	3.0	2.5
Bus	38.0	37.7	39.2	40.1
Metro	3.0	2.6	2.1	2.0
Auto-Rickshaw	44.9	45.6	45.8	45.9
Private	10.8	10.6	10.0	9.5
Auto Waiting Time Commuters (min)				
Average	21.22	22.82	27.52	24.85
Min / Max	0.01 / 46.27	0.00 / 33.90	0.02 / 30.99	0.00 / 43.10
Auto Waiting Time Passengers (min)				
Average	8.73	13.97	14.45	11.52
Min / Max	0.02 / 30.88	0.01 / 30.88	0.03 / 30.88	0.03 / 30.88
Bus Schedule Fallback				
Commuters (% of MNL-Bus)	39.9	39.2	39.1	39.3
Passengers (% of MNL-Bus)	38.0	38.8	38.1	37.7
Avg bus wait commuters (min)	11.75	10.49	10.98	11.24
Avg bus wait passengers (min)	18.76	19.23	19.18	19.65

Table 6.9: Simulation KPI Summary — Luggage Bus Coefficient Sensitivity (Fleet = 600, Weekday)

Metric	**−1.0**	**−0.4738**	**+0.4738**	**+1.0**
Demand (agents generated)				
Commuters (First Mile)	14,536	14,336	14,543	14,704
Passengers (Last Mile)	13,148	13,001	13,143	12,903
First-Mile Commuter Mode Split (%)				
Walking	3.4	3.5	3.7	3.2
Bus	33.9	35.5	37.9	39.6
Metro	8.7	8.4	6.3	5.2
Auto-Rickshaw	38.8	37.5	37.7	38.0
Private	15.2	15.0	14.4	14.0
Last-Mile Passenger Mode Split (%)				
Walking	3.3	3.2	3.1	3.0
Bus	36.8	38.0	39.6	39.6
Metro	3.9	3.0	2.0	1.8
Auto-Rickshaw	44.4	44.9	45.6	46.1
Private	11.6	10.8	9.7	9.5
Auto Waiting Time Commuters (min)				
Average	8.24	21.22	26.45	30.55
Min / Max	0.00 / 31.86	0.01 / 46.27	0.01 / 30.92	0.02 / 30.55
Auto Waiting Time Passengers (min)				
Average	16.92	8.73	14.30	16.43
Min / Max	0.03 / 30.88	0.02 / 30.88	0.02 / 30.88	0.03 / 31.03
Bus Schedule Fallback				
Commuters (% of MNL-Bus)	40.5	39.9	38.6	38.5
Passengers (% of MNL-Bus)	38.4	38.0	37.8	38.3
Avg bus wait commuters (min)	11.24	11.75	10.91	10.72
Avg bus wait passengers (min)	18.87	18.76	19.73	19.16

For waiting time co-efficient variation

The out-of-vehicle travel time coefficient (β_{OVTT}) was varied across four values (0.058, 0.1, 0.5, and 1.0) (refer table 6.10) at fleet size 600 on weekdays, where larger negative values imply greater sensitivity to waiting time. As the magnitude increases, the impact on mode split is far more pronounced than observed in either the group or luggage coefficient sensitivities. Bus share for last-mile passengers drops sharply from 38.0% at 0.058 to just 27.2% at 1.0, while private vehicle share surges from 10.8% to 23.9%, indicating that highly wait-sensitive passengers actively shift away from bus toward more reliable private options. For first-mile commuters the effect is comparatively muted, with bus share declining moderately from 35.5% to 33.8% and private vehicle share rising from 15.0% to 18.2%. Bus fallback rates decline significantly at higher magnitudes — dropping from 39.9% to 27.4% for commuters and 38.0% to 29.2% for passengers — reflecting that fewer agents choose bus through MNL when waiting time is heavily penalised. Average bus waiting times also decrease at higher magnitudes, consistent with fewer and more committed bus users remaining in the system. Overall, the OVTT coefficient is the most influential parameter tested, underestimating that perceived waiting time is the critical factor governing bus mode choice, and that reducing actual bus wait times would yield substantially greater ridership gains than adjustments to group or luggage preferences alone.

6.4.3 Demand level variation

The demand level sensitivity analysis was conducted at fleet size F=600F = 600 F=600 on weekdays by scaling the base case arrival rate to 75%, 100%, and 150%, with the 200% scenario currently incomplete and reserved for future reporting (refer table 6.11). Across the three completed scenarios, auto-rickshaw mode share remains remarkably stable for both first-mile commuters (38.9%–39.1%) and last-mile passengers (41.3%–43.5%), confirming that auto demand is largely inelastic to overall demand scaling. Bus share shows a modest declining trend for first-mile commuters as demand increases (33.3% at 100% to 32.0% at 150%), suggesting mild schedule saturation under higher loads. Auto waiting times for commuters remain broadly stable at 21–22 minutes across the three scenarios, while passenger waiting times show more variability (8.5 to 13.4 minutes), reflecting the sensitivity of the interchange idle pool to demand surges. Bus fallback rates stay within a narrow band of 38–43% across all demand levels, reinforcing the earlier finding that fallback behaviour is governed by schedule structure rather than demand volume.

Table 6.10: Simulation KPI Summary — OVTT Coefficient Sensitivity (Fleet = 600, Weekday)

Metric	**−0.058**	**−0.1**	**−0.5**	**−1.0**
Demand (agents generated)				
Commuters (First Mile)	14,336	14,851	14,483	14,093
Passengers (Last Mile)	13,001	12,657	12,691	12,260
First-Mile Commuter Mode Split (%)				
Walking	3.5	2.9	3.5	3.7
Bus	35.5	37.2	34.4	33.8
Metro	8.4	6.5	7.8	6.1
Auto-Rickshaw	37.5	38.8	37.7	38.2
Private	15.0	14.6	16.5	18.2
Last-Mile Passenger Mode Split (%)				
Walking	3.2	3.1	3.4	3.5
Bus	38.0	38.8	31.2	27.2
Metro	3.0	2.5	6.1	6.5
Auto-Rickshaw	44.9	44.9	38.9	39.0
Private	10.8	10.6	20.4	23.9
Auto Waiting Time Commuters (min)				
Average	21.22	23.63	29.56	24.56
Min / Max	0.01 / 46.27	0.01 / 30.77	0.00 / 31.72	0.01 / 28.96
Auto Waiting Time Passengers (min)				
Average	8.73	17.11	13.90	8.03
Min / Max	0.02 / 30.88	0.02 / 35.87	0.03 / 30.88	0.01 / 30.88
Bus Schedule Fallback				
Commuters (% of MNL-Bus)	39.9	39.3	34.3	27.4
Passengers (% of MNL-Bus)	38.0	37.3	35.8	29.2
Avg bus wait commuters (min)	11.75	10.85	9.27	8.68
Avg bus wait passengers (min)	18.76	19.21	18.56	18.52

Table 6.11: Simulation KPI Summary — Demand Level Sensitivity (Fleet = 600, Weekday)

Metric	**75%**	**100%**	**150%**	**200%**
Demand (agents generated)				
Commuters (First Mile)	10,659	14,047	21,295	28,474
Passengers (Last Mile)	9,114	12,529	18,631	24,979
First-Mile Commuter Mode Split (%)				
Walking	2.7	3.0	3.1	2.8
Bus	33.3	33.1	32.0	32.7
Metro	9.3	9.4	9.5	9.6
Auto-Rickshaw	38.9	38.4	39.1	38.5
Private	15.8	16.2	16.4	16.4
Last-Mile Passenger Mode Split (%)				
Walking	3.5	3.4	3.5	3.4
Bus	39.6	35.9	40.2	35.7
Metro	4.0	4.8	4.2	5.0
Auto-Rickshaw	43.0	43.5	41.3	43.2
Private	9.9	12.5	10.9	12.7
Auto Waiting Time Commuters (min)				
Average	21.18	21.95	22.36	24.58
Min / Max	0.00 / 29.89	0.00 / 29.52	0.00 / 34.32	0.00 / 43.15
Auto Waiting Time Passengers (min)				
Average	8.47	14.77	13.38	17.49
Min / Max	0.03 / 22.18	0.04 / 20.56	0.01 / 36.16	0.01 / 45.19
Bus Schedule Fallback				
Commuters (% of MNL-Bus)	40.8	41.9	42.5	41.4
Passengers (% of MNL-Bus)	38.8	38.3	36.9	38.5
Avg bus wait commuters (min)	10.86	11.35	11.24	11.36
Avg bus wait passengers (min)	20.40	19.28	20.00	19.60

Chapter 7

Conclusion and Future work

7.1 Conclusion

This study developed and demonstrated an agent-based simulation framework for assessing first- and last-mile fleet sizing strategies at a major multimodal interchange, with Yeshwantpur Railway Station, Bengaluru serving as the case study. The framework integrates real-world train ridership data, live GTFS bus schedules, metro network geometry, and a population-weighted origin–destination generation mechanism within a continuous-time AnyLogic environment. Mode choice is governed by a Multinomial Logit model that captures the heterogeneity of urban travellers across five competing modes — walking, auto-rickshaw, bus, metro, and private vehicle — as a function of socioeconomic attributes, trip characteristics, and real-time service availability.

The base case results, run across four fleet sizes ($F = 300, 400, 500, 600$) for both weekdays and weekends, reveal several consistent system-level patterns. Auto-rickshaw emerges as the dominant mode for last-mile passengers at approximately 42–43% of trips, while bus accounts for 36–40%. For first-mile commuters the split is more balanced, with auto and bus together covering around 72–74% of trips. Bus schedule fallback rates remain persistently high at 38–42% across all fleet sizes, indicating that the binding constraint on bus uptake is not traveller preference but temporal service gaps — specifically, the absence of qualifying departures within the boarding window. Auto-rickshaw waiting times show meaningful sensitivity to fleet size, ranging from approximately 6 minutes at $F = 600$ to over 24 minutes at $F = 400$ for last-mile passengers on weekdays, confirming that fleet size is a genuine lever for managing access delay. Last-mile passengers consistently experience lower auto waiting times than first-mile commuters owing to the dedicated idle pool stationed at the interchange, a structural asymmetry that the framework makes explicitly visible.

The policy case — doubling BMTC bus frequency uniformly across all routes — delivers a clear and consistent reduction in average bus waiting time of approximately 2 minutes for first-mile commuters (from 10.8–11.1 minutes to 8.4–9.0 minutes on weekdays) and 2–3 minutes for last-mile passengers (from 19–21 minutes to 17–18 minutes). However, it produces negligible improvement in bus mode share and in several scenarios slightly increases auto demand, driven by higher fallback rates among last-mile passengers who encounter more frequent buses arriving within the exclusion window at key interchange stops. This finding reveals an important asymmetry: the policy benefits first-mile commuters more reliably than last-mile passengers, be-

cause the latter face a structural 10-minute exclusion constraint at the four primary interchange bus stops that frequency alone cannot resolve.

The sensitivity analysis further establishes that the out-of-vehicle travel time coefficient (β_{OVTT}) is by far the most influential model parameter: as waiting time sensitivity increases, last-mile bus share drops sharply from 38% to 27% and private vehicle share rises correspondingly, while auto-rickshaw demand remains largely inelastic to changes in bus utility parameters across all scenarios tested. This points clearly to the conclusion that *reducing actual waiting time* — through schedule improvements, stop relocation, or fleet reallocation — will yield substantially greater ridership gains than adjustments to bus attractiveness through group or luggage preference alone.

Taken together, the framework demonstrates that no single fleet size dominates across all performance dimensions simultaneously. At $F = 600$, auto waiting times are minimised but the improvement in bus uptake is negligible, and the marginal benefit over $F = 500$ is modest. At $F = 300$, auto waiting times become prohibitively high during peak periods while bus fallback provides no relief. The results suggest that $F = 500$ represents a reasonable operating point under base case conditions for the Yeshwantpur interchange, balancing average auto waiting time (approximately 12–20 minutes across day types) against fleet utilisation, while the policy case offers the prospect of reducing required fleet size by one increment ($F = 400$) if bus frequency improvements are accompanied by targeted stop-level scheduling at the interchange proximity stops. This is the primary fleet sizing insight the framework surfaces: bus frequency policy and auto fleet sizing are interdependent levers, and optimising one without the other leaves substantial system efficiency gains unrealised.

7.2 Recommendations

Based on the simulation results, the following operational and policy recommendations are made for the Yeshwantpur interchange and, by extension, for comparable multimodal hubs in Indian cities.

- **Prioritize stop-level schedule integration over uniform frequency doubling.** The high fallback rate among last-mile passengers is driven by the absence of qualifying bus departures within a specific time window at the four key interchange bus stops (stop IDs 21288, 24016, 21131, 21132), which together account for nearly 91% of successful last-mile boardings. Targeted scheduling of additional trips from these stops during peak train arrival windows would reduce fallback more effectively than a blanket frequency increase across all routes.

- **Establish a dedicated first-mile auto allocation zone.** Unlike last-mile passengers, first-mile commuters have no idle auto pool and must wait for vehicles to arrive from within a 2 km radius. Designating a small waiting fleet at one or two high-demand residential nodes within the catchment would directly reduce the asymmetric waiting time disadvantage observed consistently across all scenarios.

- **Use fleet size $F = 500$ as the baseline operating condition.** Under current bus schedule conditions, $F = 500$ provides a reasonable balance between

auto waiting time and fleet utilisation for both weekdays and weekends. Scaling down to $F = 400$ should be considered only if bus frequency improvements are implemented simultaneously and their fallback impact on last-mile passengers is monitored.

- **Treat waiting time reduction as the primary mode shift lever.** The sensitivity analysis shows that the OVTT coefficient governs mode choice far more strongly than group size or luggage preferences. Any policy aimed at increasing bus ridership should therefore focus first on reducing actual waiting time at the boarding point rather than on service comfort or vehicle capacity adjustments.

7.3 Limitations and Future Work

Though the framework captures the essential dynamics of first- and last-mile mode choice and fleet sizing at a multimodal interchange, several modelling assumptions constrain the precision of the current results and point toward productive directions for future work.

- **MNL parameter estimation from survey data.** All MNL coefficients are adopted from related literature or course-level estimates rather than estimated from a stated or revealed preference survey at Yeshwantpur. A targeted intercept survey during train arrivals on representative weekdays and weekends would yield station-specific parameter estimates and more reliable mode share predictions.

- **Heterogeneity within group travel.** The current model assumes uniform mode choice across all members of a travelling group. Incorporating intra-group preference heterogeneity — for example, the role of a group leader or the influence of a member with heavy luggage — would more accurately represent collective decision-making.

- **Stochastic bus travel times.** Bus speeds are held constant at 20 km/h irrespective of time of day. Introducing time-varying speed profiles based on observed congestion patterns would better reflect the variability in bus journey times that commuters actually experience, particularly during peak hours.

- **Bus occupancy constraints.** The model does not impose a capacity limit on boarding buses. During peak periods, overcrowded buses may force passengers to wait for the next service, increasing effective waiting time and fallback rates beyond what is currently modelled.

- **Network distance vs Euclidean distance.** All travel time computations use straight-line Euclidean distance. Replacing this with actual route distances derived from OpenStreetMap would improve the accuracy of travel time and cost estimates, particularly for auto-rickshaw routing.

- **Train-triggered dynamic scheduling.** The policy case implements a static, uniform frequency doubling. A more operationally realistic intervention would trigger additional bus departures dynamically in response to train arrival events,

aligning supply with the episodic surge in demand that characterises a railway interchange. This represents the most natural extension of the current framework toward a fully demand-responsive scheduling model.

- **Include sub-urban in the choice set.** We can further extend the study by extending the choice set for FMLM commuters by including the upcoming sub-urban (both Corridor 1 and Corridor 2 (Rites 2019)) which will pass through Yeshwantpur interchange area and can take away a significant portion of the FMLM demand.

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